

Chapter 3

Method (additions)

3.6 Evaluation Protocol for Automatic Prompting

All automatic-prompting experiments (Section 4.3) share one protocol, so that every row of every table is directly comparable. The protocol fixes the frozen model, the data split, the training recipe of the learned prompters, the conversion of a prompter output into SAM prompts and the evaluation metrics.

Frozen model. Phase 2 uses one frozen Phase-1 model, the Infused-Token ViT-B checkpoint with MLP ratio 1.0 (Table 4.12: TC 0.724 / AR 0.632 IoU with ground-truth boxes). The Infused-Token checkpoint with MLP ratio 0.5 serves as a robustness check. Input adapter, encoder and decoder are not changed, except in the decoder-adaptation experiments (Section 3.8).

Data split. The 398 ClimateNet training images are split once into 358 training and 40 validation images (fixed random permutation). The validation images are used only to select the checkpoint of each learned prompter and to tune post-processing parameters; the 61 test images are used once per final model. (The original Phase-2 scripts selected the best epoch on the test set, which biases their numbers upwards.)

Feature cache. Because the encoder is frozen and no augmentation is used (Section 2.2.2), the encoder is run once per image and its outputs are cached: the neck embedding $E \in \mathbb{R}^{256 \times 64 \times 64}$, the outputs of all twelve ViT blocks $F_1, \dots, F_{12} \in \mathbb{R}^{768 \times 64 \times 64}$ and the four CG-Net input fields. This is numerically identical to running the encoder in every training step and reduces a 60-epoch training of a small prompter from hours to minutes.

Training of learned prompters. Every learned prompter outputs two logit maps $\ell_{\text{TC}}, \ell_{\text{AR}}$ (independent sigmoids) and is trained with the loss selected in Section 4.1 (Table 4.4), summed over both classes,

$$\mathcal{L}_{\text{seg}} = \sum_{c \in \{\text{TC}, \text{AR}\}} \left[1 - \frac{TP_c + 1}{TP_c + \alpha_c FP_c + \beta_c FN_c + 1} - \frac{1}{|\Omega|} \sum_{i \in \Omega} \alpha_{t,c} (1 - p_{t,i})^{\gamma_c} \log p_{t,i} \right], \quad (3.1)$$

with soft counts $TP_c = \sum_i p_i y_i$, $FP_c = \sum_i p_i (1 - y_i)$, $FN_c = \sum_i (1 - p_i) y_i$ per image; TC: $\alpha = 0.3$, $\beta = 0.7$, focal $\alpha = 0.95$, $\gamma = 5$; AR: $\alpha = \beta = 0.5$, focal $\alpha = 0.85$, $\gamma = 5$. Logits are upsampled bilinearly to 768×1152 before the loss. The multi-scale fusion models add their deep-supervision terms (Eq. 3.4). All models use AdamW (learning rate 10^{-3} , weight decay 10^{-4}), cosine decay to 10^{-5} , 60 epochs, an effective batch size of 8 and bf16 autocast; the checkpoint with the best validation mean foreground IoU (evaluated every second epoch) is kept. Every configuration is trained with three seeds.

From a prompter output to SAM prompts. A binary mask $m_c = \mathbb{I}[\ell_c > 0]$ is split into 8-connected components of at least 20 pixels, as in `PromptMaker`. Seven conversions are compared (Table 3.1). Boxes and points are mapped from the 1152×768 grid to SAM’s 1024×1024 frame (the image is stretched, not padded), and the masks of all prompts of a class are merged by a pixel-wise maximum of the logits, as in `ClimateSAM.assemble_raw_masks`.

Conversion	Prompt(s) per connected component
box	tight bounding box
box (+10%)	box enlarged by 10% of its width and height
points	5 positive points in the eroded component + its centroid, 5 negative points in a ring around it
box + points	both of the above
mask logits	one dense prompt per class: ℓ_c resized to 256×256 , clamped to $[-20, 20]$ (hard masks: ± 10)
box + mask	every box together with the dense prompt of its class
hybrid	TC: box + mask logits; AR: mask logits only

Table 3.1: Conversions of a prompter output into SAM prompts. The hybrid conversion follows from the prompt analysis in Section 4.3.1: dense prompts carry the shape of an AR, but a TC covers only a few cells of the prompt encoder’s 64×64 grid and needs a box.

Metrics. As in the other thesis tables, IoU is computed per class on counts summed over all $N = 61$ test images,

$$\text{IoU}_c = \frac{\sum_n TP_{c,n}}{\sum_n (TP_{c,n} + FP_{c,n} + FN_{c,n})}, \quad \text{Mean FG IoU} = \frac{1}{2} (\text{IoU}_{\text{TC}} + \text{IoU}_{\text{AR}}), \quad (3.2)$$

where a pixel predicted as both classes counts as AR. Three further analyses are used:

- **Object level.** A ground-truth object (8-connected, ≥ 20 px) is *found* if any predicted pixel overlaps it; a predicted object is *correct* if it overlaps any ground-truth object. Recall = found / ground-truth objects, precision = correct / predicted objects.
- **Error decomposition.** The IoU of a prediction is recomputed after removing one error type at a time: deleting predicted objects that touch no ground truth (*no false objects*), adding the missed ground-truth objects (*add missed objects*), both (*perfect detection*), or replacing every detected ground-truth object by its exact shape while false objects are kept (*perfect shape*). The gain of each variant measures the IoU lost to that error type.
- **Paired bootstrap.** Two outputs A and B are compared by resampling the test images with replacement (2000 times), recomputing both dataset-level IoUs on each resample, and reporting the 2.5 and 97.5 percentiles of $\text{IoU}_B - \text{IoU}_A$ as a 95% confidence interval. For learned prompters the per-image counts of the three seeds are pooled, so the interval reflects the sampling of test images, which dominates the variance (seed standard deviations are 0.001–0.01).

3.7 Mask-Prompt Generator

The multi-scale fusion generator (Figure 3.9) upsamples all twelve ViT block outputs to 1024×1024 with nearest-neighbour interpolation before it predicts. The mask-prompt generator (MPG) proposed here starts from three observations: (1) the encoder describes the image on a 64×64 grid, so upsampling its features before the prediction adds computation but no information; (2) the only output resolution SAM needs is its dense-prompt resolution, 256×256 ; (3) ARs are long, thin structures, so the prediction needs a large receptive field.

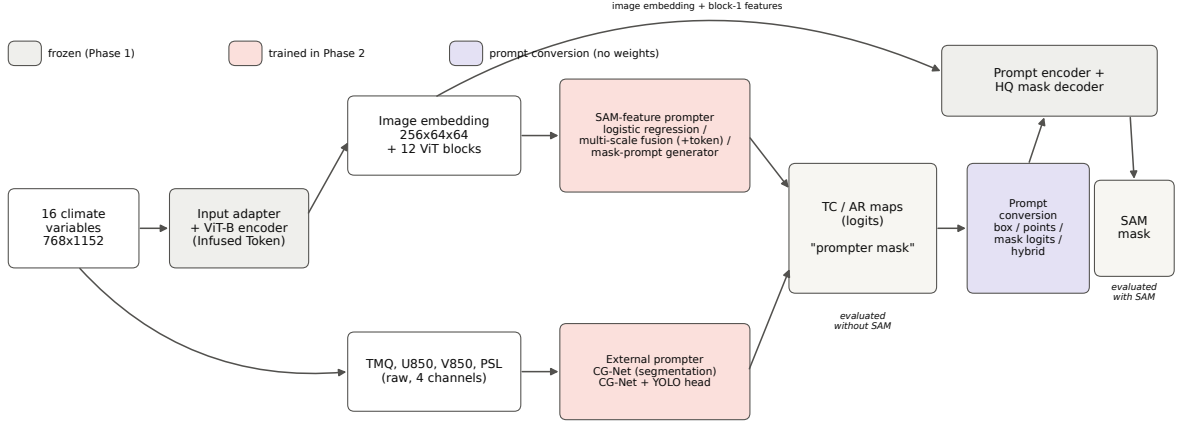


Figure 3.1: Benchmark pipeline for automatic prompting. Every prompter produces TC and AR logit maps (the *prompter mask*), which are evaluated directly and after conversion into prompts for the frozen prompt encoder and HQ decoder. Grey: frozen Phase-1 components; red: trained in Phase 2; violet: prompt conversion (no weights).

MPG projects the neck embedding E and the concatenated block outputs $[F_6; F_{12}]$ to 128 channels each by 1×1 convolutions, adds them, and applies three ConvNeXt blocks at 64×64 with dilations 1, 2 and 4 (depth-wise 3×3 convolution, LayerNorm, 1×1 expansion by 4, GELU, 1×1 projection, residual connection). Two transposed convolutions upsample to 256×256 , and a 3×3 convolution predicts the two logit maps (Figure 3.2). The bias of the last layer is initialised to the logit of a 5% foreground prior, $b = -\log \frac{1-\pi}{\pi}$ with $\pi = 0.05$, so that the initial prompts are nearly empty instead of random. The model has 0.67 M parameters and needs 5.7 GFLOPs per image, compared with 1.41 M parameters and 702 GFLOPs for the multi-scale fusion generator.

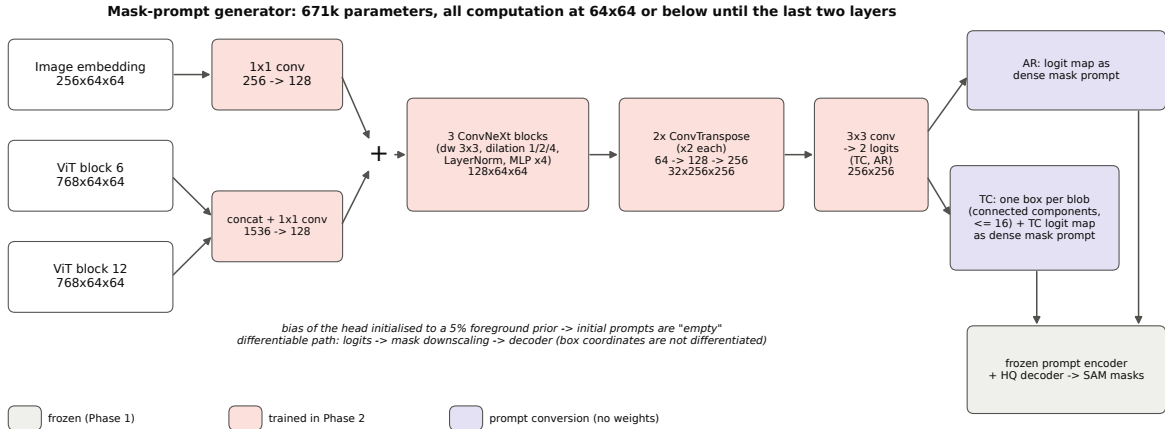


Figure 3.2: Mask-prompt generator and the hybrid conversion of its logit maps into SAM prompts.

Training through SAM. The hybrid conversion is differentiable with respect to the logit maps: the AR map enters the prompt encoder’s mask-downscaling convolutions directly, and the TC map does so together with (non-differentiable) boxes around its connected components.

MPG can therefore also be trained with a loss on SAM’s output,

$$\mathcal{L} = \lambda \mathcal{L}_{\text{seg}}(\ell, y) + \mathcal{L}_{\text{seg}}(\text{SAM}(\text{hybrid}(\ell)), y), \quad (3.3)$$

either from scratch (*end-to-end*, $\lambda = 0.5$) or starting from the model trained with $\mathcal{L}_{\text{seg}}$ alone for 30 further epochs at learning rate 10^{-4} (*two-stage*, $\lambda = 1$). The frozen decoder acts as a fixed differentiable function, so the gradient reaches the prompter (Figure 3.3).

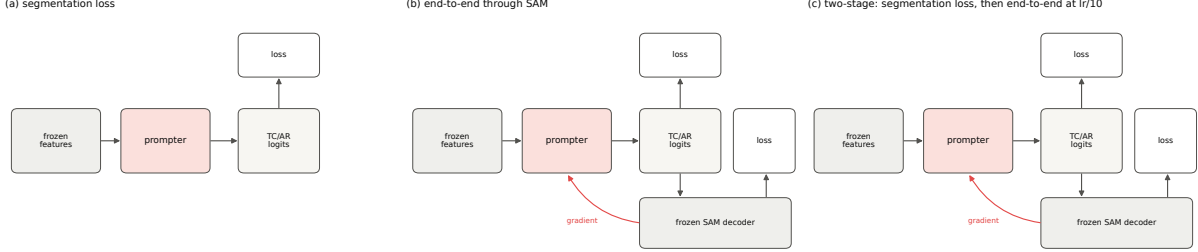


Figure 3.3: The three training modes of the mask-prompt generator: segmentation loss only, end-to-end through the frozen SAM decoder, and two-stage.

3.8 Adapting the Decoder to Generated Prompts

The Phase-1 decoder was trained only with ground-truth prompts. Two ways of exposing it to realistic prompts are tested. In both, the trainable parts of the HQ decoder (`hf_mlp_ar/tc`, `compress_vit_feat`, `embedding_encoder`, `embedding_maskfeature`; 1.21 M parameters) are fine-tuned with the loss of Eq. (3.1) on the union of the prompted masks of a class against the complete ground truth of that class, while encoder and prompt encoder stay frozen.

Adaptation to a prompter. The decoder is trained on the prompts of a fixed prompter (30 epochs, learning rate $3 \cdot 10^{-4}$). A prompter is much better on the images it was trained on than on new ones, so in-sample prompts are unrealistically clean. The *out-of-fold* variant avoids this: five copies of MPG are trained, each without one fifth of the training images, and every training image receives its prompts from the copy that did not see it.

Prompt-robust decoder. The decoder is trained on corrupted ground-truth prompts mixed with out-of-fold MPG prompts (probability 0.5 each; 25 epochs, learning rate $3 \cdot 10^{-4}$). The corruptions mimic the errors of real prompters. Every object is left without a prompt with probability 0.15; every box side is moved by $U(-0.15, 0.3)$ times the box size; Binomial(3, 0.25) false boxes of realistic size are added per class; dense prompts are eroded or dilated with an elliptical kernel of up to 5×5 px and, with probability 0.3, receive a false blob (a real object moved to a random position). The target of a box is the *complete* ground-truth object(s) it touches, and an empty mask for a box that touches no object. Unlike in Phase 1, the decoder is thus trained to output nothing for a false prompt and to extend or shrink a prompt towards the object. Two prompt modes are trained (box, hybrid), and a control variant is trained on corrupted ground-truth prompts only.

Chapter 4

Results

4.3 Automatic Prompt Generator for SAM

Phase 1 showed that the adapted SAM segments TCs and ARs well when it receives ground-truth prompts. An operational system must obtain its prompts automatically, from a *prompter*. This section asks whether such a pipeline (prompter $\rightarrow$ prompts $\rightarrow$ SAM) outperforms the prompter alone, and which prompter and prompt format work best. It is organised along five research questions:

- RQ1** How well can the frozen Phase-1 SAM segment TCs and ARs if the prompts are perfect, and how sensitive is it to realistic prompt errors? (Section 4.3.1)
- RQ2** Does SAM prompted by CG-Net improve on CG-Net, as reported in the earlier version of Table 4.13? (Section 4.3.2)
- RQ3** Which prompter produces the best masks and prompts under identical conditions, and at what cost? (Section 4.3.3)
- RQ4** Does SAM add value to any prompter, and can this be achieved by training the prompter through SAM or by adapting the decoder? (Sections 4.3.3 and 4.3.5)
- RQ5** Where exactly do the automatic systems lose IoU? (Section 4.3.4)

All experiments follow the protocol of Section 3.6: one frozen Phase-1 model (Infused Token, MLP ratio 1.0), checkpoint selection on 40 held-out training images, three seeds per learned prompter, and evaluation on the 61 test images.

4.3.1 Prompt Analysis

Before any prompter is trained, prompts built from the ground truth show what the frozen SAM can achieve and how it reacts to the errors a real prompter makes.

Prompt types. With perfect prompts the tight box is the best sparse prompt (TC 0.723 / AR 0.625), points are the weakest (0.527 / 0.577), and adding points to a box *lowers* the IoU (0.583 / 0.604), as in Table 4.12 (Figure 4.1). Dense prompts are better, because they carry the shape of the object: mask logits alone reach AR 0.920, and the hybrid prompt (TC: box + mask, AR: mask) reaches TC 0.872 / AR 0.920. These numbers are upper bounds, since a ground-truth mask prompt already contains the answer, but they show that the adapted encoder and decoder are not the bottleneck of the automatic pipeline.

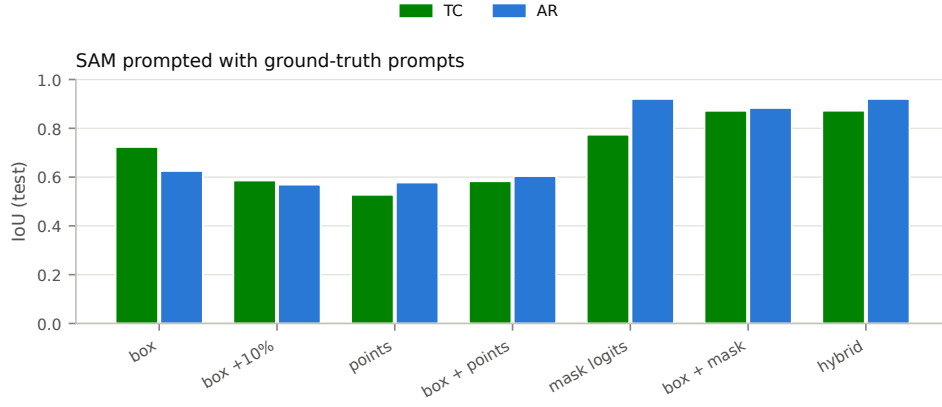


Figure 4.1: SAM prompted with prompts built from the ground truth (test set, primary checkpoint).

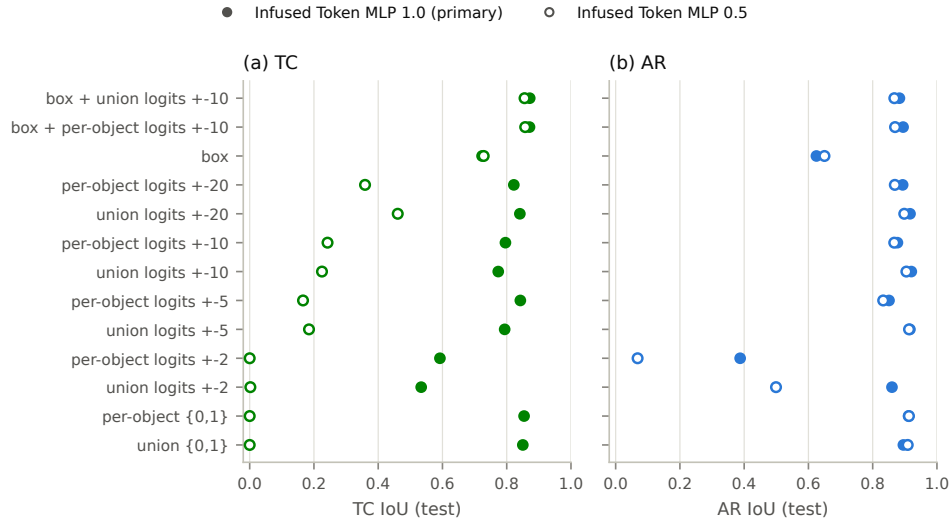


Figure 4.2: How the decoder reads dense mask prompts built from the ground truth: $\{0,1\}$ maps versus logit maps of scale $\pm a$, one prompt per class (union) versus one per object, with and without a box. Filled markers: primary checkpoint (MLP ratio 1.0); open markers: MLP ratio 0.5.

Format of dense prompts. Whether the decoder uses a dense prompt for TCs depends on the checkpoint (Figure 4.2). On the primary checkpoint both $\{0, 1\}$ and logit masks work (TC 0.85, AR 0.90); on the MLP-0.5 checkpoint a $\{0, 1\}$ mask yields no TC at all (0.000) and even ± 20 logits reach only TC 0.46, while ARs are unaffected. The prompt encoder reduces the 256×256 mask to 64×64 , where a cyclone covers only a handful of cells, and whether the decoder learned to use such weak dense evidence is a by-product of its Phase-1 training. A box together with the mask is robust on both checkpoints (TC 0.86–0.87); this motivates the hybrid conversion. One prompt per class works as well as one per object, so a single decoder call per class and image suffices for dense prompts.

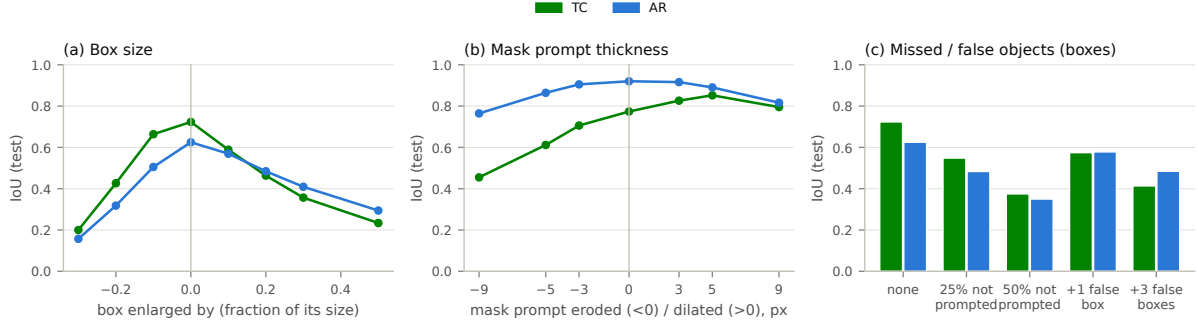


Figure 4.3: SAM with deliberately corrupted ground-truth prompts (primary checkpoint): (a) box size, (b) thickness of a dense mask prompt, (c) missed and false objects.

Ground-truth prompt	MLP 1.0	MLP 0.5	prompt-robust, hybrid
tight box	0.72 / 0.62	0.73 / 0.65	0.72 / 0.46
box enlarged 20%	0.46 / 0.48	0.44 / 0.49	0.54 / 0.48
box shrunk 20%	0.43 / 0.32	0.43 / 0.31	0.43 / 0.20
box, 25% of objects not prompted	0.55 / 0.48	0.55 / 0.50	0.54 / 0.34
box + 1 false box per class	0.57 / 0.58	0.58 / 0.58	0.60 / 0.45
box + 3 false boxes per class	0.41 / 0.48	0.41 / 0.47	0.46 / 0.41
mask logits ± 10	0.77 / 0.92	0.23 / 0.90	0.54 / 0.90
mask eroded 5 px	0.61 / 0.86	0.16 / 0.86	0.39 / 0.84
mask dilated 5 px	0.85 / 0.89	0.32 / 0.88	0.72 / 0.90
box + mask logits ± 10	0.87 / 0.88	0.86 / 0.87	0.88 / 0.83

Table 4.1: SAM prompted with prompts built from the ground truth, with controlled errors (TC IoU / AR IoU, 61 test images). Columns: the frozen Phase-1 decoders of both checkpoints and the decoders fine-tuned on generated or corrupted prompts (Section on decoder adaptation).

Sensitivity to prompt errors. Figure 4.3 and Table 4.1 quantify how SAM reacts to the errors a prompter makes:

- *Box size.* The IoU peaks sharply at the tight box. Enlarging all boxes by 20% costs 0.26 TC and 0.14 AR IoU; shrinking them by 20% costs 0.30 and 0.31. The decoder takes the extent of the box literally.
- *Missed and false objects.* An object without a prompt is never recovered (25% of the objects without a prompt: TC 0.723 $\rightarrow$ 0.548, AR 0.625 $\rightarrow$ 0.483), and one random false box per class costs 0.149 TC and 0.047 AR IoU: the decoder segments *something* inside every box it receives.

- *Dense prompts are forgiving.* Eroding the mask by 3 px costs only 0.07 TC / 0.02 AR, and dilating it by 3–5 px even *improves* TC (0.774 $\rightarrow$ 0.852), because the $4\times$ downsampling in the prompt encoder otherwise weakens small objects.

Answer to RQ1. With correct prompts the Phase-1 SAM is an excellent segmenter (hybrid prompts: TC 0.87 / AR 0.92). It does not, however, correct its prompts: missed objects stay missed, false prompts become false objects, and a box error of 10–20% already costs 0.1–0.3 IoU.

4.3.2 SAM with CG-Net as Prompter

Re-evaluation of Table 4.13. Every configuration of the earlier Table 4.13 was re-run with the thesis pipeline itself (the fine-tuned CG-Net, `PromptMaker`, `ClimateSAM.infer` and `forward`) on the 61 test images. Two image paths were compared: the path of `test_prompt_effect.py`, which calls `ClimateSAM.set_infer_img()` and thereby skips the learned input adapter (it feeds the first three raw channels to the encoder), and the training path `encode_images()`, which applies it.

Prompt type	Pos.	Neg.	Enlarge	IoU AR	IoU TC	IoU AR (orig. script)	IoU TC (orig. script)
CG-Net alone (baseline)				0.3817	0.3494		
bbox	0	0	0.0	0.3690	0.3414	0.3630	0.3390
mask	0	0	0.0	0.3758	0.3343	0.3568	0.3251
bbox	0	0	-0.1	0.3267	0.3327	0.3294	0.3315
point+bbox	15	10	0.0	0.3305	0.3252	0.3115	0.3232
point+bbox	10	5	0.0	0.3427	0.3235	0.3254	0.3181
point	5	10	0.0	0.3373	0.3178	0.3236	0.3139
point	1	3	0.0	0.2713	0.3171	0.2774	0.3138
point	5	5	0.0	0.3375	0.3128	0.3259	0.3093
bbox	0	0	0.1	0.3582	0.3073	0.3436	0.3062
point	2	2	0.0	0.3180	0.3048	0.3148	0.2986
point	1	2	0.0	0.2866	0.3038	0.2878	0.3023
point	1	1	0.0	0.3086	0.2964	0.3049	0.2946
point	10	10	0.0	0.3262	0.2923	0.3031	0.2847
bbox	0	0	0.2	0.3334	0.2703	0.3110	0.2706
bbox	0	0	-0.2	0.2288	0.2502	0.2357	0.2506
point+bbox+mask	20	10	0.0	0.3313	0.2396	0.3212	0.2184
bbox	0	0	0.3	0.2991	0.2358	0.2717	0.2376
bbox	0	0	0.5	0.2372	0.1822	0.2042	0.1837
point+bbox+mask	10	5	0.0	0.3471	0.1357	0.3390	0.1222

Table 4.2: SAM with CG-Net as prompter, re-run on the 61 test images (Infused Token MLP 1.0, Phase-1 checkpoint of Table 4.12). “orig. script”: image passed through `ClimateSAM.set_infer_img`, which skips the learned input adapter (`test_prompt_effect.py`); other columns: input adapter applied as in training. Ordered by TC IoU.

The re-run of the original script with the checkpoint of the last recorded run reproduces that run exactly (box prompts: TC 0.3404 / AR 0.3758 versus 0.3409 / 0.3758 in the stored result file), so the pipeline is faithfully reproduced. The SAM rows of the earlier Table 4.13, however, are 0.07–0.12 IoU higher than every re-run and could not be traced to any stored result; they most likely stem from an earlier state of the code. Table 4.2 and Figure 4.4 give the corrected values. The ranking of the prompt types is confirmed (tight boxes $>$ boxes + points $>$ points; enlarging boxes hurts monotonically), but **SAM prompted by CG-Net is never better than CG-Net itself**: the best configuration, tight boxes, reaches TC 0.341 / AR 0.369 against 0.349 / 0.382 for CG-Net alone. The skipped input adapter is a genuine bug but changes the IoU by only 0.00–0.03. The “mask” row depends on the checkpoint, as explained in Section 4.3.1.

CG-Net with a YOLO box head. The thesis also attached a YOLO-style detection head to CG-Net that predicts boxes directly. Its boxes are the weakest prompts of the study (Figure 4.5;

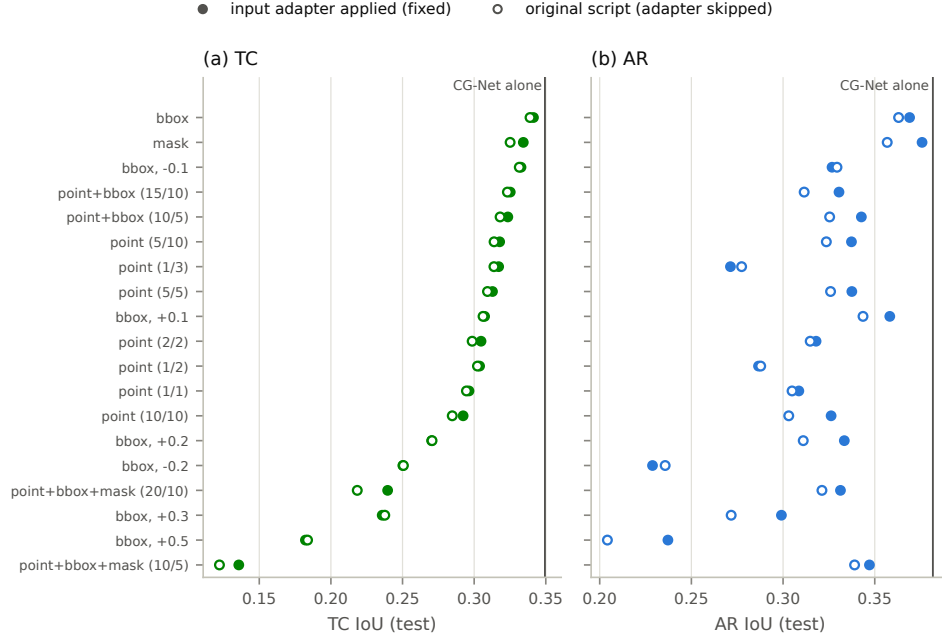


Figure 4.4: SAM with CG-Net as prompt, all configurations of Table 4.13 (primary checkpoint). Filled: input adapter applied; open: original script. Vertical line: CG-Net alone.

SAM: TC 0.19–0.22, AR 0.22–0.29 for the three stored checkpoints and all objectness thresholds), far below boxes derived from CG-Net’s segmentation (0.34 / 0.37). Directly regressed boxes are too loose for thin, curved ARs, and the prompt analysis showed that a box error of 20% alone costs 0.14–0.26 IoU.

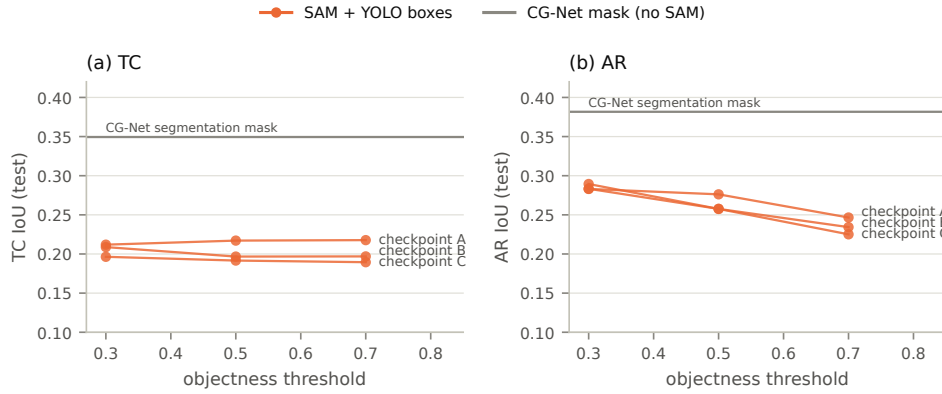


Figure 4.5: SAM prompted by the boxes of the CG-Net YOLO head (three stored checkpoints A–C) as a function of the objectness threshold. Grey line: CG-Net’s own segmentation.

Answer to RQ2. The ranking of prompt types in Table 4.13 holds, its central claim does not: with correct numbers, SAM prompted by CG-Net is slightly worse than CG-Net for every prompt type.

4.3.3 Generator from SAM Inputs

The prompts of this section read the frozen encoder’s features instead of the raw climate fields: the linear probe (logistic regression) on the first and on the last ViT block, the multi-

scale fusion generator (MSF, Figure 3.9) and its token-gated variants, and the mask-prompt generator (MPG, Section 3.7). The original multi-scale generators predicted three channels, of which the loss supervised only TC and AR as independent sigmoids while the prediction was the three-channel argmax, so an untrained background channel competed with TC and AR; here all prompts use two sigmoid channels thresholded at zero. The original logistic-regression prompter read the *first* ViT block; it is evaluated as is and on the last block.

Prompter	Output	TC IoU	AR IoU	BG IoU	Mean IoU	Mean FG IoU
Ground truth (oracle)	Prompter mask (no SAM)	1.000	1.000	1.000	1.000	1.000
	SAM + box	0.723	0.625	0.971	0.773	0.674
	SAM + hybrid	0.872	0.920	0.994	0.929	0.896
CG-Net (official weights)	Prompter mask (no SAM)	0.327	0.333	0.926	0.529	0.330
	SAM + box	0.318	0.342	0.931	0.530	0.330
	SAM + hybrid	0.327	0.339	0.929	0.532	0.333
CG-Net (fine-tuned)	Prompter mask (no SAM)	0.349	0.382	0.940	0.557	0.366
	SAM + box	0.341	0.369	0.939	0.550	0.355
	SAM + hybrid	0.349	0.384	0.941	0.558	0.367
Logistic regression, ViT block 1	Prompter mask (no SAM)	0.195±0.001	0.319±0.001	0.927±0.004	0.480±0.002	0.257±0.001
	SAM + box	0.200±0.002	0.321±0.003	0.925±0.004	0.482±0.003	0.260±0.002
	SAM + hybrid	0.213±0.000	0.316±0.002	0.939±0.001	0.489±0.001	0.264±0.001
Logistic regression, ViT block 12	Prompter mask (no SAM)	0.301±0.004	0.385±0.001	0.938±0.001	0.541±0.002	0.343±0.002
	SAM + box	0.291±0.001	0.362±0.001	0.932±0.001	0.528±0.000	0.326±0.000
	SAM + hybrid	0.304±0.004	0.373±0.002	0.946±0.000	0.541±0.002	0.338±0.002
Multi-scale fusion	Prompter mask (no SAM)	0.333±0.002	0.403±0.004	0.944±0.003	0.560±0.001	0.368±0.003
	SAM + box	0.339±0.001	0.385±0.002	0.941±0.003	0.555±0.001	0.362±0.001
	SAM + hybrid	0.330±0.002	0.388±0.010	0.949±0.001	0.556±0.003	0.359±0.005
Multi-scale fusion + token gate	Prompter mask (no SAM)	0.324±0.001	0.399±0.003	0.944±0.002	0.556±0.001	0.362±0.001
	SAM + box	0.336±0.002	0.382±0.003	0.940±0.002	0.553±0.001	0.359±0.002
	SAM + hybrid	0.329±0.004	0.387±0.006	0.949±0.001	0.555±0.003	0.358±0.005
Mask-prompt generator (segmentation loss)	Prompter mask (no SAM)	0.344±0.003	0.413±0.001	0.944±0.000	0.567±0.002	0.379±0.002
	SAM + box	0.343±0.006	0.391±0.001	0.940±0.001	0.558±0.002	0.367±0.003
	SAM + hybrid	0.342±0.004	0.398±0.002	0.949±0.000	0.563±0.002	0.370±0.002
Mask-prompt generator (end-to-end via SAM)	Prompter mask (no SAM)	0.335±0.010	0.378±0.019	0.941±0.006	0.551±0.010	0.356±0.014
	SAM + box	0.336±0.011	0.385±0.005	0.940±0.007	0.554±0.003	0.360±0.005
	SAM + hybrid	0.334±0.009	0.402±0.007	0.946±0.003	0.561±0.004	0.368±0.008
Mask-prompt generator (two-stage)	Prompter mask (no SAM)	0.338±0.005	0.409±0.002	0.938±0.002	0.562±0.000	0.374±0.001
	SAM + box	0.334±0.003	0.386±0.004	0.935±0.002	0.552±0.001	0.360±0.001
	SAM + hybrid	0.337±0.007	0.411±0.002	0.947±0.001	0.565±0.002	0.374±0.004
Mask-prompt generator (segmentation loss, label smoothing)	Prompter mask (no SAM)	0.338±0.009	0.412±0.001	0.943±0.001	0.564±0.003	0.375±0.005
	SAM + box	0.340±0.007	0.388±0.003	0.939±0.001	0.556±0.002	0.364±0.003
	SAM + hybrid	0.335±0.007	0.401±0.002	0.949±0.000	0.562±0.003	0.368±0.005
Learned static prompts (4 tokens / class)	SAM + learned static prompts	0.237±0.001	0.324±0.008	0.926±0.005	0.495±0.004	0.280±0.004
Learned static prompts (16 tokens / class)	SAM + learned static prompts	0.248	0.329	0.930	0.502	0.288

Table 4.3: Automatic prompting on the ClimateNet test set (61 images), frozen ClimateSAM (Infused Token, MLP ratio 1.0). Learned prompts: mean±std over 3 seeds, checkpoint selected on 40 held-out training images.

Prompters. Table 4.3 and Figure 4.6 compare all prompters. The mask-prompt generator gives the best prompter mask (TC 0.344 ± 0.003 , AR 0.413 ± 0.001). The paired bootstrap (Table 4.4) locates its advantage over the fine-tuned CG-Net entirely in the ARs (AR $+0.031$, 95% CI $[+0.020, +0.043]$; TC -0.005 , $[-0.030, +0.020]$); it is also better than MSF ($+0.011$ mean FG IoU, $[+0.003, +0.019]$) and than the linear probe on the last block ($+0.036$, $[+0.025, +0.046]$). The token gate *lowers* MSF slightly but significantly (-0.006 , $[-0.009, -0.003]$): its inputs are the fixed HQ tokens of the frozen decoder, so the gate is a learned per-class channel weighting that carries no image-dependent information. Label smoothing, which helped in Phase 1, does not help the prompter (-0.004 , $[-0.007, -0.000]$). A linear probe on the last ViT block (1.5k parameters) already reaches TC 0.301 / AR 0.385, close to the fine-tuned CG-Net, while the same probe on the first block, the original configuration, reaches only 0.195 / 0.319: the late features of the adapted encoder are almost linearly separable with respect to TC and AR, the early ones are not.

Added value of SAM. Figure 4.7 answers the central question of Phase 2. Point prompts always lose (-0.03 to -0.08 mean FG IoU). Tight boxes are roughly neutral for TCs (-0.010 to $+0.011$) and lose 0.013–0.024 AR IoU for every prompter with a good AR mask; they are

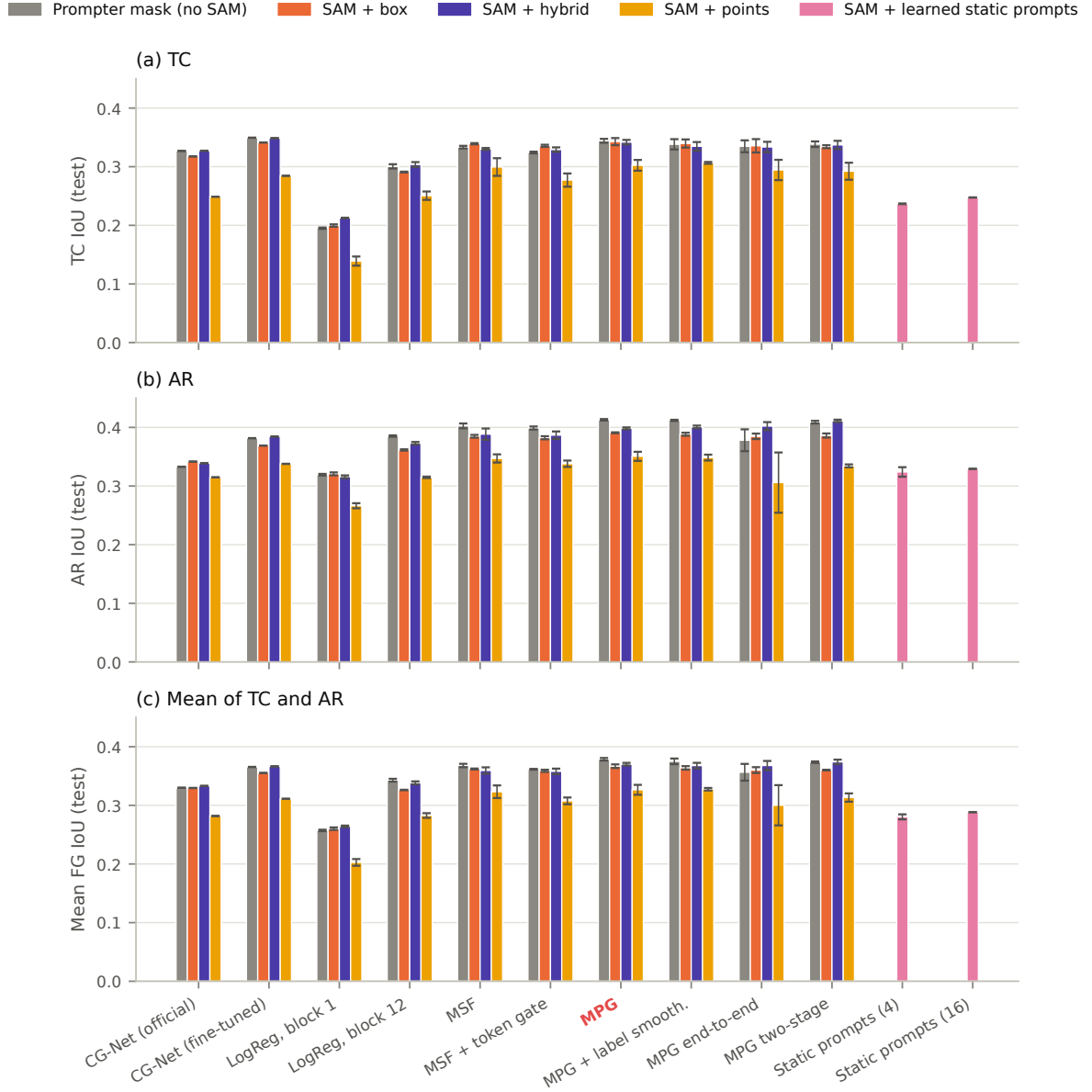


Figure 4.6: Prompter mask (grey) versus SAM prompted by it with boxes (orange), hybrid prompts (violet) and points (yellow); learned static prompts (magenta) have no prompter. Error bars: standard deviation over three seeds.

Comparison (B – A)	Δ TC [95% CI]	Δ AR [95% CI]	Δ mean FG [95% CI]
MPG vs. CG-Net (fine-tuned), masks	-0.005 [-0.030, +0.020]	+0.031 [+0.020, +0.043]	+0.013 [-0.000, +0.027]
MPG vs. MSF, masks	+0.011 [-0.003, +0.025]	+0.010 [+0.005, +0.016]	+0.011 [+0.003, +0.019]
MPG vs. LogReg block 12, masks	+0.044 [+0.025, +0.062]	+0.028 [+0.020, +0.036]	+0.036 [+0.025, +0.046]
MSF + token gate vs. MSF, masks	-0.009 [-0.015, -0.003]	-0.004 [-0.005, -0.002]	-0.006 [-0.009, -0.003]
MPG: SAM + box vs. its mask	-0.002 [-0.009, +0.006]	-0.022 [-0.028, -0.016]	-0.012 [-0.017, -0.007]
MPG: SAM + hybrid vs. its mask	-0.002 [-0.007, +0.003]	-0.015 [-0.024, -0.006]	-0.008 [-0.014, -0.003]
CG-Net (ft.): SAM + box vs. its mask	-0.008 [-0.017, +0.001]	-0.013 [-0.018, -0.006]	-0.010 [-0.016, -0.004]
MPG end-to-end: SAM + hybrid vs. its mask	-0.001 [-0.005, +0.003]	+0.024 [+0.021, +0.028]	+0.012 [+0.009, +0.015]
MPG two-stage: SAM + hybrid vs. MPG mask	-0.007 [-0.013, -0.000]	-0.002 [-0.007, +0.003]	-0.004 [-0.008, -0.000]

Table 4.4: Selected paired-bootstrap comparisons on the test set (2000 resamples of the 61 images, seeds pooled). The full list is given in Table A.2.

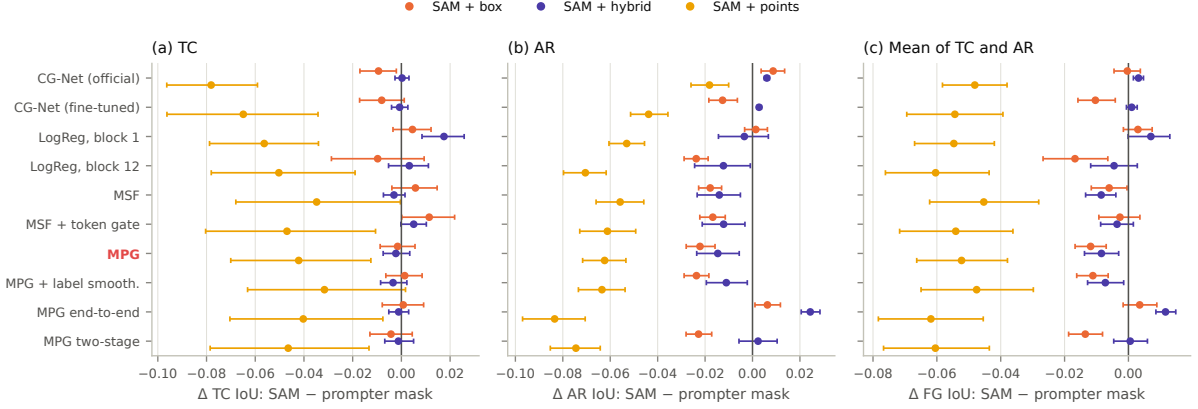


Figure 4.7: Effect of SAM: IoU of SAM’s output minus IoU of the prompter’s own mask, per prompt and prompt type, with paired-bootstrap 95% intervals.

neutral or slightly positive only where the prompter’s AR mask is poor (official CG-Net, linear probe on block 1, end-to-end MPG). The hybrid prompt is the best conversion: within about ± 0.01 of the prompter mask for every prompt, slightly *positive* for prompters with a poor mask (linear probe on block 1: $+0.007$; end-to-end MPG: $+0.012$) and slightly *negative* for the strong ones (MSF -0.009 , MPG -0.008). The best SAM output of the study (two-stage MPG, hybrid prompts, 0.374 mean FG IoU) does not exceed the best prompter mask (MPG, 0.379). The complete comparison of all seven conversions is given in Table A.1.

Training through SAM. Training MPG end-to-end through the frozen decoder makes *its own* mask worse (0.356 versus 0.379) and SAM’s output better than that mask ($+0.012$): the prompter learns to produce logits that SAM turns into good masks. The resulting SAM output (0.368) nevertheless stays below the plain MPG mask (-0.011 , $[-0.016, -0.005]$), and the end-to-end mask is strongly fragmented (Table 4.6), because a loss on SAM’s output does not penalise speckle that SAM ignores. The two-stage variant keeps the MPG mask quality (0.374) and gives the best SAM output of the study (0.374), again not above the MPG mask (-0.004 , $[-0.008, -0.000]$). Its selected validation epochs are 2–8 of 30: once the prompter has been trained on its own maps, training through SAM barely changes it.

Model	Parameters	GFLOPs / image	Latency (ms, A100, batch 1)
CG-Net	494,232	22.93	8.89
Logistic regression	1,538	0.01	0.05
Multi-scale fusion	1,406,474	702.36	36.13
Multi-scale fusion + token gate	1,415,180	702.89	38.12
Mask-prompt generator	671,138	5.74	0.89
(frozen ClimateSAM encoder, for reference)	95,987,456	973.25	122.39
(one SAM decoder call, one box, for reference)	5,269,508	16.80	3.39

Table 4.5: Cost of the prompts on top of the frozen ClimateSAM encoder (forward pass, one image; FLOPs counted with `torch.utils.flop_counter`, latency = median of 20 runs). The encoder and one decoder call are listed for reference.

Cost. MPG is both the most accurate and one of the cheapest prompters (Figure 4.8, Table 4.5): 5.7 GFLOPs and 0.9 ms per image, i.e. $122\times$ fewer FLOPs and $40\times$ lower latency than MSF and $4\times$ fewer FLOPs than CG-Net. Relative to the frozen encoder (973 GFLOPs) the prompter adds 0.6%. MSF spends almost all of its computation on 3×3 convolutions at 1024×1024 over features that were upsampled by nearest-neighbour interpolation from 64×64 ,

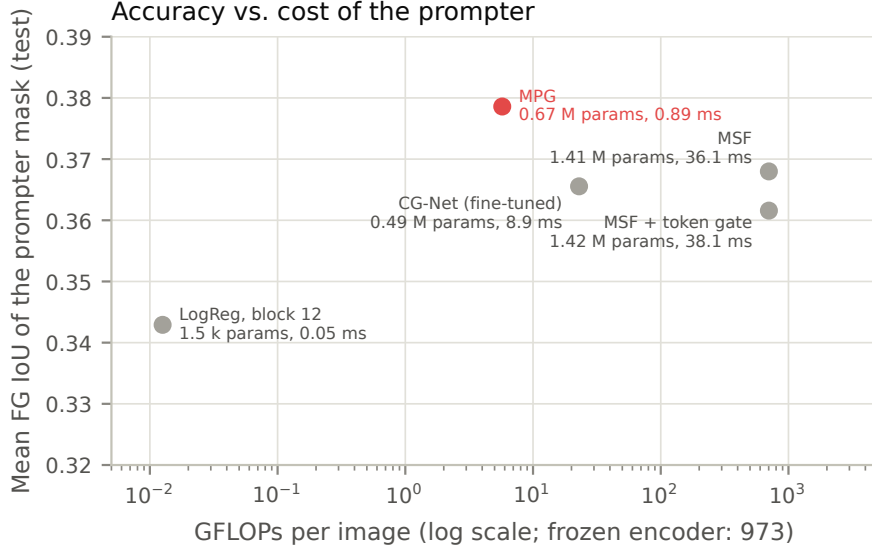


Figure 4.8: Mean FG IoU of the prompter mask versus forward cost per image (log scale).

i.e. on a resolution that contains no additional information. With cached features a complete MPG training takes about 4 minutes and converges within 10–15 epochs (Figure 4.9); an MSF training takes 46–83 minutes.

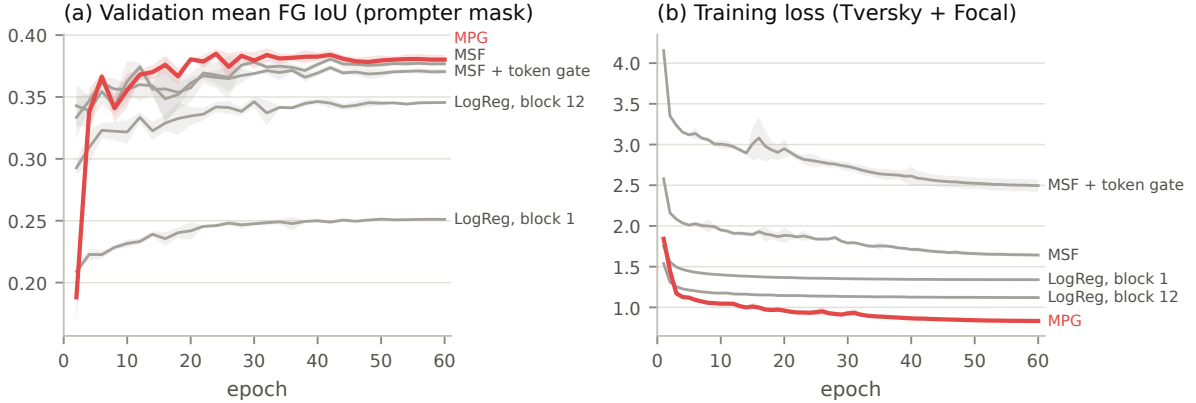


Figure 4.9: Validation mean FG IoU of the prompter mask and training loss (mean $\pm$ std over three seeds). Loss values are not comparable across architectures, because the MSF models add deep-supervision terms.

Answer to RQ3 and RQ4. The best prompter is the small mask-prompt generator. SAM prompted by any prompter is at best as good as that prompter, and training the prompter through SAM does not change this.

4.3.4 Error Analysis

Detection. All prompters find more than 90% of the ARs but only about two thirds of the TCs, and 40–60% of their predicted objects touch no ground-truth object (Table 4.6, Figure 4.10). For ARs many of these are small fragments: the prompters predict 12–18 AR objects per image, against 6.8 in the ground truth.

Prompter	TC recall	TC precision	AR recall	AR precision	TC obj./img	AR obj./img
Ground truth (oracle)	1.000	1.000	1.000	1.000	2.9	6.8
CG-Net (official weights)	0.726	0.411	0.935	0.571	5.1	11.1
CG-Net (fine-tuned)	0.682	0.513	0.908	0.567	3.9	11.9
Logistic regression, ViT block 1	0.767±0.006	0.265±0.016	0.967±0.004	0.324±0.018	8.8	25.2
Logistic regression, ViT block 12	0.739±0.012	0.419±0.018	0.966±0.004	0.418±0.010	5.1	17.6
Multi-scale fusion	0.670±0.030	0.529±0.061	0.956±0.001	0.482±0.017	4.0	17.4
Multi-scale fusion + token gate	0.670±0.020	0.448±0.016	0.960±0.005	0.444±0.025	4.7	20.3
Mask-prompt generator (segmentation loss)	0.661±0.009	0.563±0.031	0.951±0.001	0.514±0.024	3.5	14.9
Mask-prompt generator (end-to-end via SAM)	0.652±0.012	0.521±0.050	0.963±0.007	0.362±0.107	3.7	90.0
Mask-prompt generator (two-stage)	0.667±0.006	0.546±0.039	0.962±0.003	0.430±0.008	3.7	18.0
Mask-prompt generator (segmentation loss, label smoothing)	0.648±0.015	0.558±0.026	0.953±0.010	0.495±0.007	3.5	15.2

Table 4.6: Object-level detection of the prompter masks: a ground-truth object is found if any predicted pixel overlaps it; a predicted object is correct if it overlaps any ground-truth object. Objects are 8-connected components of at least 20 pixels; with this counting the test set has 2.9 TC and 6.8 AR objects per image.

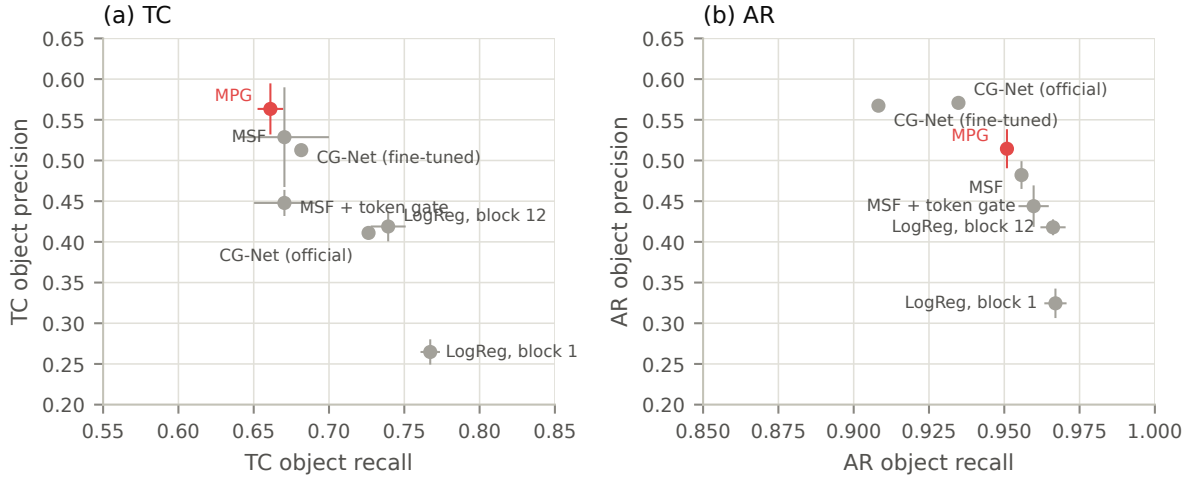


Figure 4.10: Object-level recall and precision of the prompter masks (mean over seeds; bars: standard deviation).

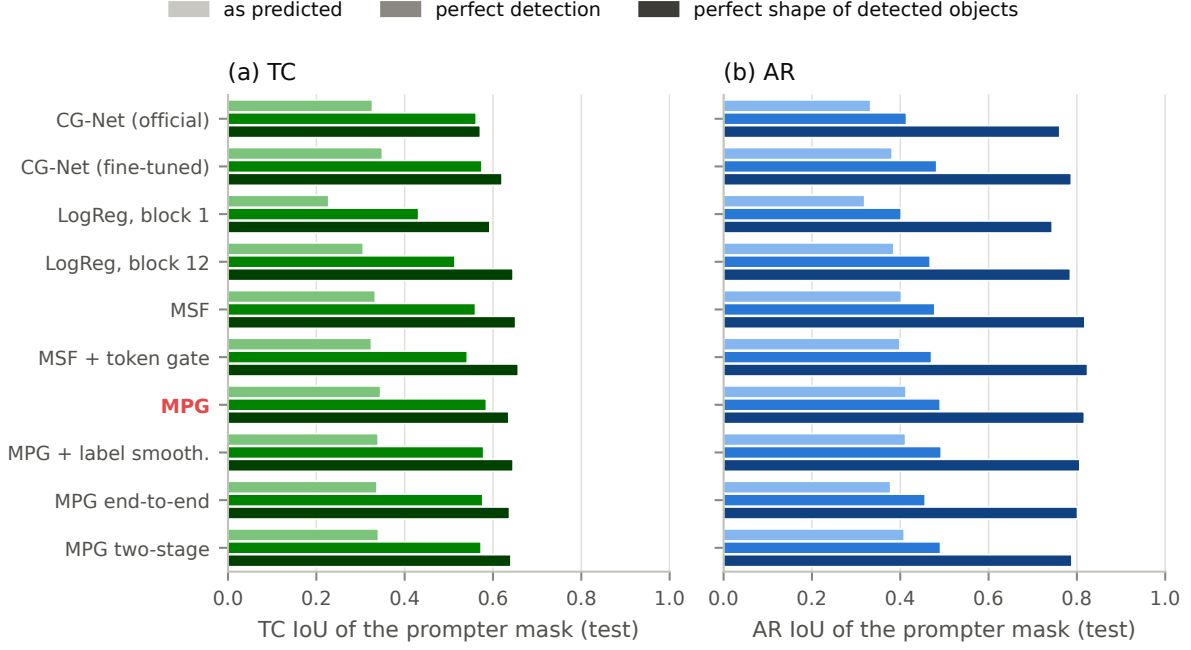


Figure 4.11: IoU of the prompter masks as predicted, with perfect detection (false objects removed, missed objects added), and with the true shape of every detected object (false objects kept).

Detection versus delineation. The error decomposition (Figure 4.11, Table A.3) separates the two error sources.

- *ARs*. For MPG, perfect detection raises the AR IoU only from 0.413 to 0.490, but giving every *detected* river its true shape raises it to 0.817. The rivers are found; what is wrong is their extent, i.e. their length, their width and where a river ends.
- *TCs*. Perfect detection (0.345 $\rightarrow$ 0.585) and perfect shapes of the detected cyclones ($\rightarrow$ 0.636) matter about equally: a third of the cyclones is missed, half of the predicted blobs are false, and the small objects are also delineated imprecisely.

These are exactly the errors that SAM cannot repair (Section 4.3.1): missed objects stay missed, false prompts become false objects, and the extent of an AR is taken from the prompt. This is the mechanism behind the result of Section 4.3.3.

Answer to RQ5. ARs are detected ($> 90\%$) but their extent is wrong; TCs are missed (one third) and falsely detected (about half of the predicted blobs).

4.3.5 Adapting the Decoder to Generated Prompts

Adaptation to a prompter. Fine-tuning the decoder on generated prompts (Section 3.8) teaches it something useful: with out-of-fold MPG boxes it rejects part of the false AR prompts (AR object precision 0.50 $\rightarrow$ 0.62) and it recovers most of what SAM loses relative to the prompter (Figure 4.12, Table 4.7). In no configuration, however, does SAM end up above the prompter mask (best: -0.004 mean FG IoU with out-of-fold hybrid prompts); on the second checkpoint the best configuration ties the prompter ($+0.001$).

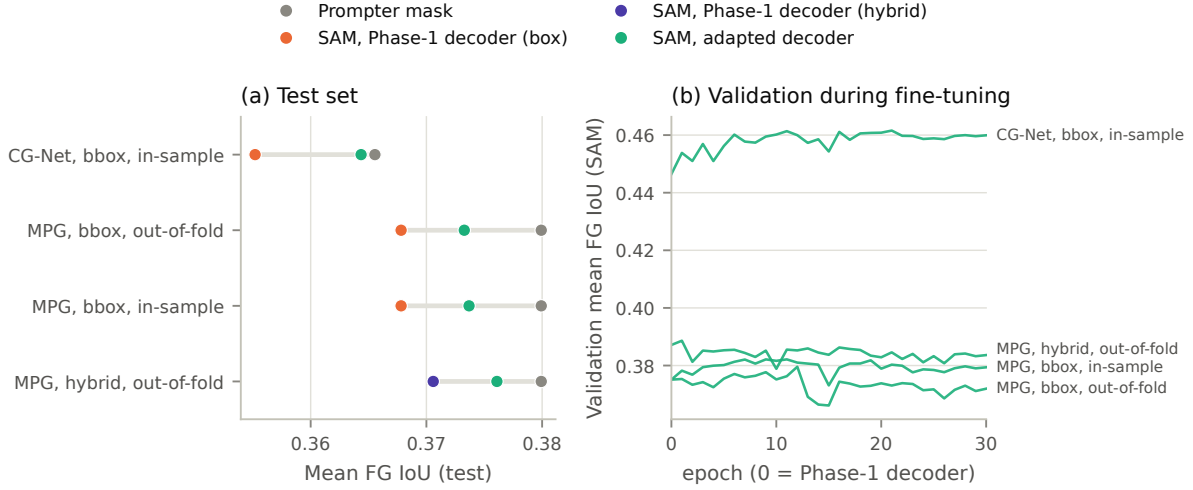


Figure 4.12: (a) Test mean FG IoU of the prompter mask, SAM with the Phase-1 decoder and SAM with the adapted decoder; (b) validation IoU during fine-tuning (epoch 0: Phase-1 decoder; the CG-Net curve is high because CG-Net was trained on these images).

Prompter, prompts, training prompts	Output	TC IoU	AR IoU	Mean FG IoU	TC recall / prec.	AR recall / prec.
CG-Net (fine-tuned), bbox, in-sample	prompter mask	0.349	0.382	0.366		
	SAM, Phase-1 decoder	0.341	0.369	0.355	0.67 / 0.56	0.90 / 0.53
	SAM, adapted decoder	0.345	0.384	0.364	0.64 / 0.58	0.88 / 0.58
Mask-prompt generator, bbox, out-of-fold	prompter mask	0.346	0.414	0.380		
	SAM, Phase-1 decoder	0.344	0.392	0.368	0.65 / 0.59	0.95 / 0.50
	SAM, adapted decoder	0.342	0.404	0.373	0.65 / 0.57	0.93 / 0.62
Mask-prompt generator, bbox, in-sample	prompter mask	0.346	0.414	0.380		
	SAM, Phase-1 decoder	0.344	0.392	0.368	0.65 / 0.59	0.95 / 0.50
	SAM, adapted decoder	0.343	0.404	0.374	0.65 / 0.59	0.95 / 0.55
Mask-prompt generator, hybrid, out-of-fold	prompter mask	0.346	0.414	0.380		
	SAM, Phase-1 decoder	0.344	0.397	0.371	0.65 / 0.59	0.91 / 0.65
	SAM, adapted decoder	0.347	0.405	0.376	0.65 / 0.60	0.94 / 0.64

Table 4.7: Adapting the HQ decoder parts to generated prompts (test set). Phase-1 decoder = the frozen decoder of all other experiments; adapted = best validation epoch of 30.

4.3.6 Evaluation-only Improvements

4.3.7 Prompt-free SAM with Learned Static Prompts

The most minimal prompt-free variant needs no prompter: K sparse prompt tokens per class (initialised from SAM’s positive-point embedding) are learned and fed to the frozen decoder for every image, so all image-specific information must come from the decoder’s attention to the image embedding. With $K = 4$ this reaches TC 0.237 ± 0.001 / AR 0.324 ± 0.008 (three seeds), with $K = 16$ TC 0.248 / AR 0.329, i.e. 0.09–0.10 mean FG IoU below MPG. Object recall is high (TC 0.87, AR 0.98), but the output breaks into thousands of fragments (object precision about 0.2): a fixed query cannot decide *which* humid, windy region is a TC or an AR. This decision needs an image-dependent prompter.

4.3.8 Robustness to the Frozen Checkpoint

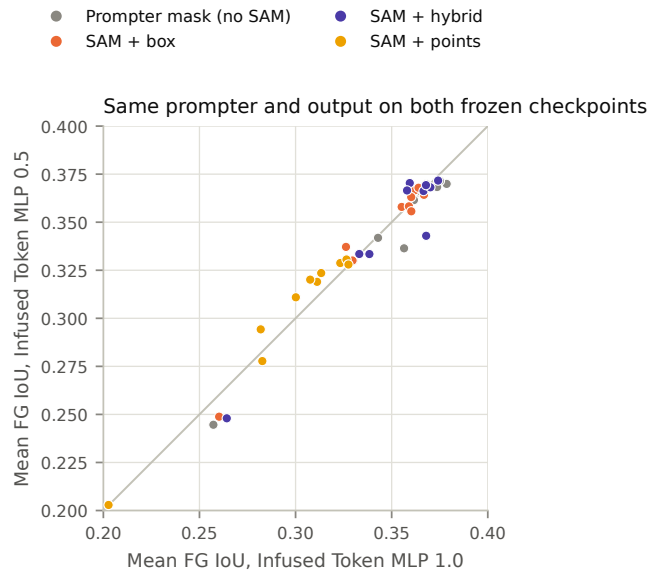


Figure 4.13: Mean FG IoU of every prompter and output on the MLP-1.0 (x) and MLP-0.5 (y) checkpoints. Diagonal: equal performance.

All experiments were repeated with the MLP-0.5 checkpoint (Table A.5; MSF models with one seed). The results lie close to the diagonal (Figure 4.13), so the conclusions do not depend on the checkpoint. On this checkpoint MPG (0.370) ties MSF (0.370) and the smoothed MPG (0.371); all three stay ahead of CG-Net (0.366; AR +0.027 to +0.028) and of the linear probe (0.342), and again no SAM output exceeds the best prompter mask (best SAM output 0.373 versus 0.371, within noise).

4.3.9 Qualitative Examples

Figure 4.14 illustrates the quantitative findings. All prompters find the long rivers of the southern hemisphere and the large cyclone over the Gulf of Mexico; they differ in the extent of the rivers and in small false or missed objects. SAM with boxes (e, g) fills every box with a river-shaped mask, including the false AR over the central North Pacific that the prompters hallucinated, and cuts rivers at the box borders; the hybrid prompt (f) follows the prompter’s shape closely. The learned static prompts (h) mark every moist band. A second example is shown in Figure A.1 (Appendix).

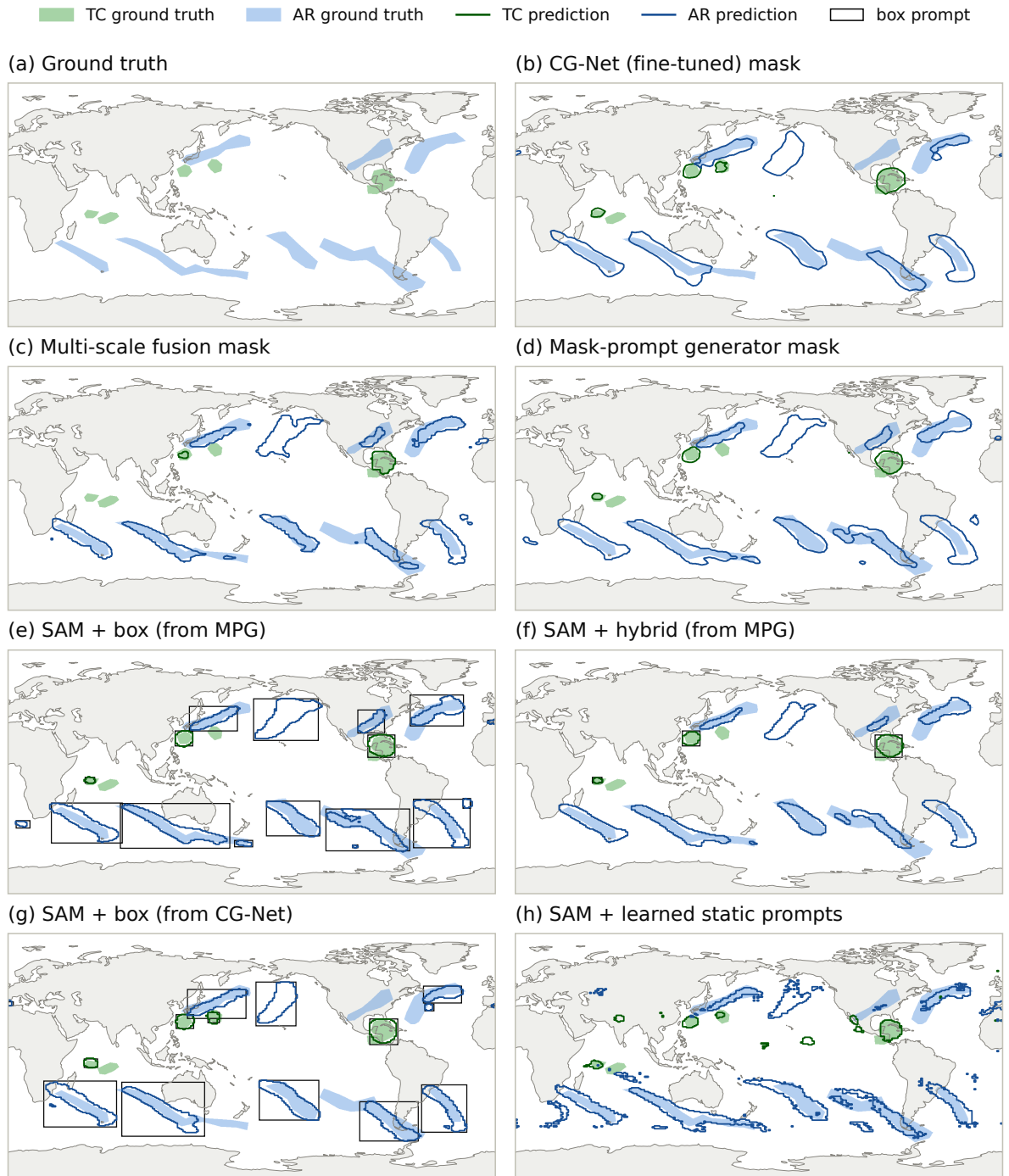


Figure 4.14: Test image 17. Filled: ground truth (TC green, AR blue); outlines: prediction; black rectangles: box prompts. (b–d) prompter masks; (e, f) SAM prompted by MPG with boxes and with hybrid prompts; (g) SAM prompted by CG-Net boxes; (h) SAM with learned static prompts.

4.3.10 Summary

- **RQ1.** With perfect prompts the Phase-1 SAM reaches TC 0.87 / AR 0.92 (hybrid prompts). It does not correct prompts: missed objects stay missed, false prompts become false objects, and 10–20% box error costs 0.1–0.3 IoU.
- **RQ2.** The SAM rows of the earlier Table 4.13 do not reproduce; SAM prompted by CG-Net is slightly worse than CG-Net for every prompt type. The ranking of the prompt types is confirmed.
- **RQ3.** The mask-prompt generator (0.67 M parameters, 5.7 GFLOPs) is the best prompter: TC 0.344 / AR 0.413, AR +0.031 over the fine-tuned CG-Net and +0.011 mean FG IoU over the multi-scale fusion generator at $122\times$ fewer FLOPs.
- **RQ4.** SAM prompted by any automatic prompter is at best as good as that prompter. Neither training the prompter through SAM nor adapting the decoder to realistic prompts changes this.
- **RQ5.** ARs are found but their extent is wrong; TCs are missed and falsely detected. A prompt-conditioned decoder cannot repair either error.

Chapter 5

Discussion

5.1 Discussion of Automatic Prompting

5.1.1 Why SAM Cannot Improve on Its Prompter

The central finding of Section 4.3 is that SAM with automatic prompts is bounded by its prompter. This follows from what the Phase-1 decoder was trained to do. It learned $p(\text{mask} \mid \text{image}, \text{prompt})$ from prompts that were always correct: every box contained exactly one object, and the extent of the box was the extent of the object. Under this training distribution the optimal decoder *trusts* its prompt: it segments whatever lies inside a box and never has a reason to output nothing or to extend a mask beyond its prompt. The prompt analysis confirms this behaviour directly (Figure 4.3). The errors of a prompter are of exactly these kinds, namely false objects, missed objects and wrong AR extents (Figures 4.10 and 4.11), so they pass through SAM unchanged. What SAM can add is boundary quality inside a correct prompt; but for ARs the dominant error is the extent of the whole river, not its boundary pixels, and for TCs the objects are small enough that detection dominates.

The decoder-adaptation experiments show that this is not only a question of the training distribution. Even a decoder trained on realistic, out-of-fold prompts learns to reject some false prompts but cannot add information that the prompt does not contain. Prompter and decoder read the same frozen image embedding; whatever the decoder could infer about the extent of an AR from that embedding, a prompter trained directly on the labels infers as well.

A further reason lies in the labels. ClimateNet’s labels are the consensus of several experts, and the mean performance of a single expert against this consensus is AR 0.341 / TC 0.257 IoU (Table 2.1). All good prompters of this study (TC 0.33–0.35, AR 0.38–0.41) already exceed the average single expert. The remaining errors, such as the extent of a river, where it starts and ends, or weak cyclones, are precisely where experts disagree, so part of the gap to the oracle is irreducible label ambiguity rather than a deficiency of the model.

5.1.2 Why a Small Generator Beats the Larger Ones

The encoder describes the image on a 64×64 grid of 16×16 -pixel patches. The ground truth has a resolution of 768×1152 , but its information content at object boundaries is limited by the ambiguity between experts. MSF upsamples the features to 1024×1024 by nearest-neighbour interpolation and runs most of its 702 GFLOPs there, which cannot create information and makes the model slow to train. MPG spends its capacity where the information is, at 64×64 , with a receptive field of about 15 patches from the dilated convolutions on top of the encoder’s own global attention, and uses two learned upsampling steps only to reach the 256×256 resolution of SAM’s dense prompt. The strong linear probe on the last ViT block supports this view: the adapted encoder has already done most of the work, and the prompter mainly aggregates context and removes noise.

5.1.3 Lessons for Prompt Design

1. If sparse prompts are used, they should be tight boxes derived from a segmentation. Points, loose boxes and boxes regressed by a detector are harmful.
2. Dense prompts are more robust to prompt errors and carry the shape of an AR, but they must be passed as logits in the value range the decoder was trained with, and small TCs need an accompanying box on checkpoints whose decoder ignores weak dense evidence.
3. One dense prompt per class is sufficient. This makes SAM decoding cheap: one decoder call per class and image instead of one per object.

5.1.4 Implications for a Prompt-free ClimateSAM

For a prompt-free system, the prompter’s own mask is the natural output (MPG: TC 0.344 / AR 0.413, mean IoU 0.567 including background). SAM with hybrid prompts gives masks of statistically similar quality (-0.008 mean FG IoU) with smoother boundaries, at the cost of two decoder calls per image. The contribution of the foundation model is thus its *encoder*: the adapted features allow a 0.67M-parameter head, and even a linear probe, to match or exceed a dedicated segmentation network trained on the raw fields. The promptable decoder remains valuable for interactive use, where a human annotator provides the boxes (Table 4.12: TC 0.72 / AR 0.63–0.64 with ground-truth boxes), but not as a refinement stage after an automatic prompter.

5.1.5 Limitations

- *Small test set.* With 61 test images, differences below about 0.01 IoU lie within the bootstrap intervals and should not be interpreted. Checkpoints are selected on only 40 validation images drawn from the same years as the training images.
- *Seeds.* Three seeds per configuration on the primary checkpoint; on the second checkpoint the MSF models were trained with one seed. The seed variance (standard deviation 0.001–0.01) is small compared with the variance from sampling test images.
- *Two frozen checkpoints.* Both are Infused-Token ViT-B models; LoRA-adapted and ViT-L encoders were not tested in Phase 2.
- *External prompter.* The fine-tuned CG-Net was trained on all 398 training images, including the validation images of this protocol; this affects only experiments that use its outputs on training images.
- *Object metrics.* An object counts as found if one pixel overlaps; stricter matching (IoU ≥ 0.5) would lower all recalls and precisions but not change the comparison.
- *Earlier Table 4.13.* The re-run shows that its SAM rows do not reproduce with the current code and checkpoints; their origin could not be identified.

5.1.6 Future Work

1. *Train Phase 1 with realistic prompts.* Mixing ground-truth prompts with corrupted ones, including false boxes whose target is empty, and with prompts of an out-of-fold prompter already during Phase 1 attacks the root cause identified above. The prompt analysis of Section 4.3.1 provides the test: a better decoder flattens the curves of Figure 4.3.

2. *Query-based prompt-free decoding.* Replacing static prompts by image-dependent object queries (DETR/Mask2Former style) with a matching-based loss turns SAM into an instance segmenter and addresses the TC detection errors directly.
3. *Joint fine-tuning of encoder adapters and prompter.* Unfreezing the infused-token adapters together with MPG on the segmentation loss lets the encoder adapt to the prompt-free objective.
4. *Longitudinal periodicity.* Circular padding in the prompter and test-time averaging over longitudinally rolled inputs would fix objects cut at the date line.
5. *Stricter object metrics.* Object recall and precision at $\text{IoU} \geq 0.3 / 0.5$ and event counts per image would make the detection analysis comparable with the ClimateNet literature.

Appendix A

Automatic Prompting: Additional Material

A.1 Complete Results of the Automatic-Prompting Benchmark

Prompter	Prompter mask (no SAM)	SAM + box	SAM + box (+10%)	SAM + points	SAM + box + points	SAM + mask logits	SAM + box + mask	SAM + hybrid	mean(prompter, SAM hybrid)	SAM + learned static prompts
Ground truth (oracle)	1.00 / 1.00	0.72 / 0.62	0.59 / 0.57	0.53 / 0.58	0.58 / 0.60	0.77 / 0.92	0.87 / 0.88	0.87 / 0.92	0.95 / 0.98	–
CG-Net (official weights)	0.33 / 0.33	0.32 / 0.34	0.28 / 0.34	0.25 / 0.31	0.28 / 0.32	0.33 / 0.34	0.33 / 0.34	0.33 / 0.34	–	–
CG-Net (fine-tuned)	0.35 / 0.38	0.34 / 0.37	0.31 / 0.36	0.28 / 0.34	0.32 / 0.36	0.34 / 0.38	0.35 / 0.38	0.35 / 0.38	–	–
Logistic regression, ViT block 1	0.20 / 0.32	0.20 / 0.32	0.19 / 0.32	0.14 / 0.27	0.18 / 0.30	0.04 / 0.32	0.20 / 0.31	0.21 / 0.32	0.21 / 0.32	–
Logistic regression, ViT block 12	0.30 / 0.39	0.29 / 0.36	0.28 / 0.35	0.25 / 0.31	0.28 / 0.35	0.04 / 0.37	0.29 / 0.36	0.30 / 0.37	0.30 / 0.37	–
Multi-scale fusion	0.33 / 0.40	0.34 / 0.38	0.33 / 0.38	0.30 / 0.35	0.33 / 0.37	0.31 / 0.39	0.33 / 0.38	0.33 / 0.39	0.33 / 0.39	–
Multi-scale fusion + token gate	0.32 / 0.40	0.34 / 0.38	0.32 / 0.38	0.28 / 0.34	0.32 / 0.37	0.26 / 0.39	0.33 / 0.38	0.33 / 0.39	0.33 / 0.39	–
Mask-prompt generator (segmentation loss)	0.34 / 0.41	0.34 / 0.39	0.32 / 0.38	0.30 / 0.35	0.32 / 0.37	0.30 / 0.40	0.34 / 0.39	0.34 / 0.40	0.34 / 0.40	–
Mask-prompt generator (end-to-end via SAM)	0.34 / 0.38	0.34 / 0.38	0.32 / 0.38	0.29 / 0.31	0.32 / 0.36	0.30 / 0.40	0.33 / 0.38	0.33 / 0.40	0.33 / 0.40	–
Mask-prompt generator (two-stage)	0.34 / 0.41	0.33 / 0.39	0.31 / 0.36	0.29 / 0.33	0.32 / 0.36	0.29 / 0.41	0.33 / 0.40	0.34 / 0.41	0.34 / 0.41	–
Mask-prompt generator (segmentation loss, label smoothing)	0.34 / 0.41	0.34 / 0.39	0.32 / 0.37	0.31 / 0.35	0.33 / 0.37	0.29 / 0.40	0.33 / 0.40	0.33 / 0.40	0.34 / 0.40	–
Learned static prompts (4 tokens / class)	–	–	–	–	–	–	–	–	–	0.24 / 0.32
Learned static prompts (16 tokens / class)	–	–	–	–	–	–	–	–	–	0.25 / 0.33

Table A.1: TC IoU / AR IoU for every way of turning the prompter output into SAM prompts (Infused Token, MLP ratio 1.0).

Comparison (B minus A)	Δ TC IoU [95% CI]	Δ AR IoU [95% CI]	Δ mean FG IoU [95% CI]	P(Δ FG>0)
CG-Net (official weights): SAM + box vs. prompter mask	-0.009 [-0.017, -0.002]	+0.009 [+0.004, +0.014]	-0.000 [-0.005, +0.004]	0.45
CG-Net (official weights): SAM + hybrid vs. prompter mask	+0.000 [-0.003, +0.003]	+0.006 [+0.005, +0.007]	+0.003 [+0.002, +0.005]	1.00
CG-Net (fine-tuned): SAM + box vs. prompter mask	-0.008 [-0.017, +0.001]	-0.013 [-0.018, -0.006]	-0.010 [-0.016, -0.004]	0.00
CG-Net (fine-tuned): SAM + hybrid vs. prompter mask	-0.001 [-0.004, +0.003]	+0.003 [+0.002, +0.004]	+0.001 [-0.001, +0.003]	0.89
Logistic regression, ViT block 1: SAM + box vs. prompter mask	+0.005 [-0.003, +0.012]	+0.001 [-0.003, +0.006]	+0.003 [-0.002, +0.007]	0.90
Logistic regression, ViT block 1: SAM + hybrid vs. prompter mask	+0.017 [+0.008, +0.026]	-0.003 [-0.014, +0.007]	+0.007 [-0.000, +0.013]	0.97
Logistic regression, ViT block 1: mean(prompter, SAM hybrid) vs. prompter mask	+0.017 [+0.009, +0.025]	-0.003 [-0.013, +0.007]	+0.007 [+0.001, +0.013]	0.98
Logistic regression, ViT block 12: SAM + box vs. prompter mask	-0.010 [-0.029, +0.009]	-0.024 [-0.029, -0.019]	-0.017 [-0.027, -0.006]	0.00
Logistic regression, ViT block 12: SAM + hybrid vs. prompter mask	+0.003 [-0.005, +0.011]	-0.012 [-0.024, -0.001]	-0.005 [-0.012, +0.003]	0.11
Logistic regression, ViT block 12: mean(prompter, SAM hybrid) vs. prompter mask	+0.004 [-0.004, +0.011]	-0.011 [-0.023, +0.000]	-0.004 [-0.011, +0.003]	0.15
Multi-scale fusion: SAM + box vs. prompter mask	+0.006 [-0.004, +0.015]	-0.018 [-0.023, -0.013]	-0.006 [-0.012, -0.000]	0.01
Multi-scale fusion: SAM + hybrid vs. prompter mask	-0.003 [-0.007, +0.001]	-0.014 [-0.023, -0.005]	-0.009 [-0.013, -0.004]	0.00
Multi-scale fusion: mean(prompter, SAM hybrid) vs. prompter mask	-0.000 [-0.003, +0.003]	-0.012 [-0.020, -0.004]	-0.006 [-0.010, -0.002]	0.00
Multi-scale fusion + token gate: SAM + box vs. prompter mask	+0.011 [+0.000, +0.022]	-0.017 [-0.022, -0.011]	-0.003 [-0.009, +0.004]	0.22
Multi-scale fusion + token gate: SAM + hybrid vs. prompter mask	+0.005 [-0.000, +0.010]	-0.012 [-0.021, -0.003]	-0.004 [-0.009, +0.002]	0.09
Multi-scale fusion + token gate: mean(prompter, SAM hybrid) vs. prompter mask	+0.005 [+0.002, +0.009]	-0.010 [-0.018, -0.002]	-0.002 [-0.007, +0.002]	0.14
Mask-prompt generator (segmentation loss): SAM + box vs. prompter mask	-0.002 [-0.009, +0.006]	-0.022 [-0.028, -0.016]	-0.012 [-0.017, -0.007]	0.00
Mask-prompt generator (segmentation loss): SAM + hybrid vs. prompter mask	-0.002 [-0.007, +0.003]	-0.015 [-0.024, -0.006]	-0.008 [-0.014, -0.003]	0.00
Mask-prompt generator (segmentation loss): mean(prompter, SAM hybrid) vs. prompter mask	-0.000 [-0.005, +0.005]	-0.013 [-0.021, -0.004]	-0.006 [-0.011, -0.001]	0.01
Mask-prompt generator (end-to-end via SAM): SAM + box vs. prompter mask	+0.001 [-0.008, +0.009]	+0.006 [+0.001, +0.012]	+0.004 [-0.002, +0.009]	0.91
Mask-prompt generator (end-to-end via SAM): SAM + hybrid vs. prompter mask	-0.001 [-0.005, +0.003]	+0.024 [+0.021, +0.028]	+0.012 [+0.009, +0.015]	1.00
Mask-prompt generator (end-to-end via SAM): mean(prompter, SAM hybrid) vs. prompter mask	-0.000 [-0.004, +0.003]	+0.024 [+0.020, +0.028]	+0.012 [+0.009, +0.015]	1.00
Mask-prompt generator (two-stage): SAM + box vs. prompter mask	-0.004 [-0.013, +0.004]	-0.023 [-0.028, -0.017]	-0.014 [-0.019, -0.008]	0.00
Mask-prompt generator (two-stage): SAM + hybrid vs. prompter mask	-0.001 [-0.007, +0.005]	+0.002 [-0.006, +0.010]	+0.001 [-0.005, +0.006]	0.58
Mask-prompt generator (two-stage): mean(prompter, SAM hybrid) vs. prompter mask	+0.001 [-0.004, +0.006]	+0.003 [-0.004, +0.011]	+0.002 [-0.003, +0.007]	0.80
Mask-prompt generator (segmentation loss, label smoothing): SAM + box vs. prompter mask	+0.001 [-0.006, +0.009]	-0.024 [-0.029, -0.018]	-0.011 [-0.016, -0.006]	0.00
Mask-prompt generator (segmentation loss, label smoothing): SAM + hybrid vs. prompter mask	-0.003 [-0.009, +0.002]	-0.011 [-0.019, -0.002]	-0.007 [-0.013, -0.001]	0.01
Mask-prompt generator (segmentation loss, label smoothing): mean(prompter, SAM hybrid) vs. prompter mask	-0.001 [-0.005, +0.004]	-0.009 [-0.017, -0.001]	-0.005 [-0.010, -0.000]	0.02
Mask-prompt generator (segmentation loss) vs. CG-Net (fine-tuned) (prompter masks)	-0.005 [-0.030, +0.020]	+0.031 [+0.020, +0.043]	+0.013 [-0.000, +0.027]	0.97
Mask-prompt generator (segmentation loss) vs. Multi-scale fusion (prompter masks)	+0.011 [-0.003, +0.025]	+0.010 [+0.005, +0.016]	+0.011 [+0.003, +0.019]	0.99
Mask-prompt generator (segmentation loss) vs. Logistic regression, ViT block 12 (prompter masks)	+0.044 [+0.025, +0.062]	+0.028 [+0.020, +0.036]	+0.036 [+0.025, +0.046]	1.00
Multi-scale fusion + token gate vs. Multi-scale fusion (prompter masks)	-0.009 [-0.015, -0.003]	-0.004 [-0.005, -0.002]	-0.006 [-0.009, -0.003]	0.00
Logistic regression, ViT block 12 vs. Logistic regression, ViT block 1 (prompter masks)	+0.105 [+0.078, +0.133]	+0.066 [+0.053, +0.080]	+0.086 [+0.070, +0.101]	1.00
Mask-prompt generator (segmentation loss, label smoothing) vs. Mask-prompt generator (segmentation loss) (prompter masks)	-0.006 [-0.012, +0.000]	-0.001 [-0.004, +0.001]	-0.004 [-0.007, -0.000]	0.01
Multi-scale fusion vs. CG-Net (fine-tuned) (prompter masks)	-0.016 [-0.041, +0.008]	+0.021 [+0.009, +0.033]	+0.003 [-0.011, +0.016]	0.64
Mask-prompt generator (two-stage): SAM + hybrid vs. segmentation-loss generator mask	-0.007 [-0.013, -0.000]	-0.002 [-0.007, +0.003]	-0.004 [-0.008, -0.000]	0.02
Mask-prompt generator (end-to-end via SAM): SAM + hybrid vs. segmentation-loss generator mask	-0.011 [-0.020, -0.001]	-0.010 [-0.018, -0.003]	-0.011 [-0.016, -0.005]	0.00
Learned static prompts (4 tokens / class) vs. segmentation-loss generator mask	-0.107 [-0.142, -0.073]	-0.089 [-0.102, -0.077]	-0.098 [-0.117, -0.079]	0.00
Learned static prompts (16 tokens / class) vs. segmentation-loss generator mask	-0.097 [-0.131, -0.063]	-0.084 [-0.095, -0.072]	-0.090 [-0.110, -0.071]	0.00

Table A.2: Paired bootstrap over the 61 test images (2000 resamples; learned prompters: the three seeds pooled). Infused Token, MLP ratio 1.0.

Prompter	Class	As is	No false objects	Add missed objects	Perfect detection	Perfect shape
CG-Net (official weights)	TC	0.327	0.453	0.405	0.562	0.571
CG-Net (fine-tuned)	TC	0.349	0.439	0.457	0.575	0.620
Logistic regression, ViT block 1	TC	0.228±0.000	0.324±0.008	0.304±0.005	0.432±0.004	0.593±0.015
Logistic regression, ViT block 12	TC	0.306±0.003	0.393±0.012	0.401±0.007	0.514±0.002	0.645±0.012
Multi-scale fusion	TC	0.333±0.002	0.405±0.011	0.461±0.018	0.560±0.011	0.651±0.005
Multi-scale fusion + token gate	TC	0.325±0.001	0.395±0.004	0.446±0.009	0.542±0.008	0.657±0.005
Mask-prompt generator (segmentation loss)	TC	0.345±0.005	0.420±0.002	0.481±0.015	0.585±0.009	0.636±0.012
Mask-prompt generator (end-to-end via SAM)	TC	0.337±0.008	0.410±0.005	0.474±0.012	0.577±0.010	0.637±0.019
Mask-prompt generator (two-stage)	TC	0.340±0.005	0.415±0.013	0.470±0.003	0.573±0.014	0.641±0.019
Mask-prompt generator (segmentation loss, label smoothing)	TC	0.340±0.010	0.407±0.011	0.483±0.016	0.579±0.015	0.645±0.002
CG-Net (official weights)	AR	0.333	0.396	0.348	0.414	0.761
CG-Net (fine-tuned)	AR	0.382	0.437	0.421	0.482	0.787
Logistic regression, ViT block 1	AR	0.319±0.001	0.396±0.004	0.324±0.003	0.402±0.002	0.744±0.021
Logistic regression, ViT block 12	AR	0.385±0.001	0.463±0.002	0.390±0.002	0.468±0.002	0.785±0.004
Multi-scale fusion	AR	0.403±0.004	0.468±0.009	0.411±0.005	0.478±0.011	0.818±0.017
Multi-scale fusion + token gate	AR	0.399±0.003	0.462±0.009	0.407±0.001	0.471±0.007	0.824±0.016
Mask-prompt generator (segmentation loss)	AR	0.413±0.001	0.479±0.002	0.423±0.001	0.490±0.002	0.817±0.004
Mask-prompt generator (end-to-end via SAM)	AR	0.378±0.019	0.451±0.018	0.383±0.020	0.456±0.017	0.801±0.053
Mask-prompt generator (two-stage)	AR	0.409±0.002	0.485±0.003	0.415±0.003	0.491±0.002	0.789±0.013
Mask-prompt generator (segmentation loss, label smoothing)	AR	0.412±0.001	0.483±0.006	0.421±0.003	0.493±0.003	0.806±0.008

Table A.3: IoU of the prompter mask after removing one error type: deleting predicted objects that touch no ground truth, adding the ground-truth objects that were missed, both, or giving every detected object its exact ground-truth shape.

Prompter	Trainable parameters	Training time (min)	Selected epoch
Logistic regression, ViT block 1	1,538	1.4	46
Logistic regression, ViT block 12	1,538	2.0	39
Multi-scale fusion	1,406,474	82.6	38
Multi-scale fusion + token gate	1,415,180	46.0	39
Mask-prompt generator (segmentation loss)	671,138	3.7	29
Mask-prompt generator (end-to-end via SAM)	671,138	26.4	23
Mask-prompt generator (two-stage)	671,138	12.4	4
Mask-prompt generator (segmentation loss, label smoothing)	671,138	4.2	29
Learned static prompts (4 tokens / class)	2,048	12.6	48
Learned static prompts (16 tokens / class)	8,192	14.2	48

Table A.4: Size and training cost (single A100, cached encoder features, 60 epochs; two-stage: 30 extra epochs on top of the segmentation-loss model).

Prompter	Output	TC IoU	AR IoU	BG IoU	Mean IoU	Mean FG IoU
Ground truth (oracle)	Prompter mask (no SAM)	1.000	1.000	1.000	1.000	1.000
	SAM + box	0.728	0.650	0.973	0.784	0.689
	SAM + hybrid	0.857	0.905	0.993	0.918	0.881
CG-Net (official weights)	Prompter mask (no SAM)	0.327	0.333	0.926	0.529	0.330
	SAM + box	0.318	0.342	0.931	0.530	0.330
	SAM + hybrid	0.328	0.339	0.929	0.532	0.333
CG-Net (fine-tuned)	Prompter mask (no SAM)	0.349	0.382	0.940	0.557	0.366
	SAM + box	0.341	0.375	0.940	0.552	0.358
	SAM + hybrid	0.348	0.384	0.941	0.558	0.366
Logistic regression, ViT block 1	Prompter mask (no SAM)	0.185±0.001	0.305±0.001	0.924±0.003	0.471±0.002	0.245±0.001
	SAM + box	0.186±0.002	0.312±0.003	0.925±0.002	0.474±0.000	0.249±0.001
	SAM + hybrid	0.186±0.001	0.310±0.001	0.928±0.003	0.474±0.001	0.248±0.001
Logistic regression, ViT block 12	Prompter mask (no SAM)	0.295±0.004	0.388±0.002	0.941±0.001	0.542±0.002	0.342±0.003
	SAM + box	0.297±0.002	0.377±0.001	0.937±0.001	0.537±0.002	0.337±0.002
	SAM + hybrid	0.300±0.004	0.367±0.006	0.946±0.001	0.538±0.003	0.333±0.004
Multi-scale fusion	Prompter mask (no SAM)	0.330	0.409	0.943	0.561	0.370
	SAM + box	0.337	0.397	0.941	0.558	0.367
	SAM + hybrid	0.337	0.404	0.946	0.562	0.370
Multi-scale fusion + token gate	Prompter mask (no SAM)	0.319	0.404	0.941	0.555	0.362
	SAM + box	0.327	0.390	0.938	0.552	0.358
	SAM + hybrid	0.329	0.404	0.945	0.559	0.367
Mask-prompt generator (segmentation loss)	Prompter mask (no SAM)	0.330±0.010	0.410±0.005	0.944±0.003	0.561±0.004	0.370±0.008
	SAM + box	0.333±0.008	0.395±0.003	0.941±0.003	0.557±0.003	0.364±0.006
	SAM + hybrid	0.332±0.011	0.404±0.010	0.947±0.002	0.561±0.006	0.368±0.010
Mask-prompt generator (end-to-end via SAM)	Prompter mask (no SAM)	0.323±0.009	0.350±0.015	0.947±0.001	0.540±0.007	0.336±0.012
	SAM + box	0.328±0.008	0.383±0.005	0.943±0.003	0.552±0.003	0.356±0.006
	SAM + hybrid	0.328±0.009	0.357±0.024	0.951±0.001	0.546±0.010	0.343±0.016
Mask-prompt generator (two-stage)	Prompter mask (no SAM)	0.326±0.005	0.410±0.001	0.940±0.002	0.559±0.002	0.368±0.003
	SAM + box	0.333±0.007	0.393±0.004	0.937±0.002	0.554±0.004	0.363±0.005
	SAM + hybrid	0.331±0.007	0.413±0.002	0.945±0.001	0.563±0.002	0.372±0.003
Mask-prompt generator (segmentation loss, label smoothing)	Prompter mask (no SAM)	0.334±0.004	0.409±0.003	0.945±0.001	0.563±0.002	0.371±0.002
	SAM + box	0.338±0.008	0.398±0.002	0.943±0.001	0.560±0.003	0.368±0.004
	SAM + hybrid	0.336±0.004	0.403±0.004	0.948±0.000	0.562±0.001	0.369±0.002

Table A.5: Automatic prompting on the ClimateNet test set (61 images), frozen ClimateSAM (Infused Token, MLP ratio 0.5). Learned prompts: mean±std over 3 seeds, checkpoint selected on 40 held-out training images.

Prompter, prompts, training prompts	Output	TC IoU	AR IoU	Mean FG IoU	TC recall / prec.	AR recall / prec.
Mask-prompt generator, bbox, out-of-fold	prompter mask	0.337	0.414	0.376		
	SAM, Phase-1 decoder	0.340	0.399	0.369	0.62 / 0.55	0.95 / 0.56
	SAM, adapted decoder	0.316	0.390	0.353	0.63 / 0.52	0.91 / 0.62
Mask-prompt generator, bbox, in-sample	prompter mask	0.337	0.414	0.376		
	SAM, Phase-1 decoder	0.340	0.399	0.369	0.62 / 0.55	0.95 / 0.56
	SAM, adapted decoder	0.340	0.404	0.372	0.63 / 0.56	0.93 / 0.64
Mask-prompt generator, hybrid, out-of-fold	prompter mask	0.337	0.414	0.376		
	SAM, Phase-1 decoder	0.339	0.413	0.376	0.62 / 0.56	0.92 / 0.63
	SAM, adapted decoder	0.338	0.416	0.377	0.63 / 0.55	0.94 / 0.62

Table A.6: Adapting the HQ decoder parts to generated prompts (test set). Phase-1 decoder = the frozen decoder of all other experiments; adapted = best validation epoch of 30.

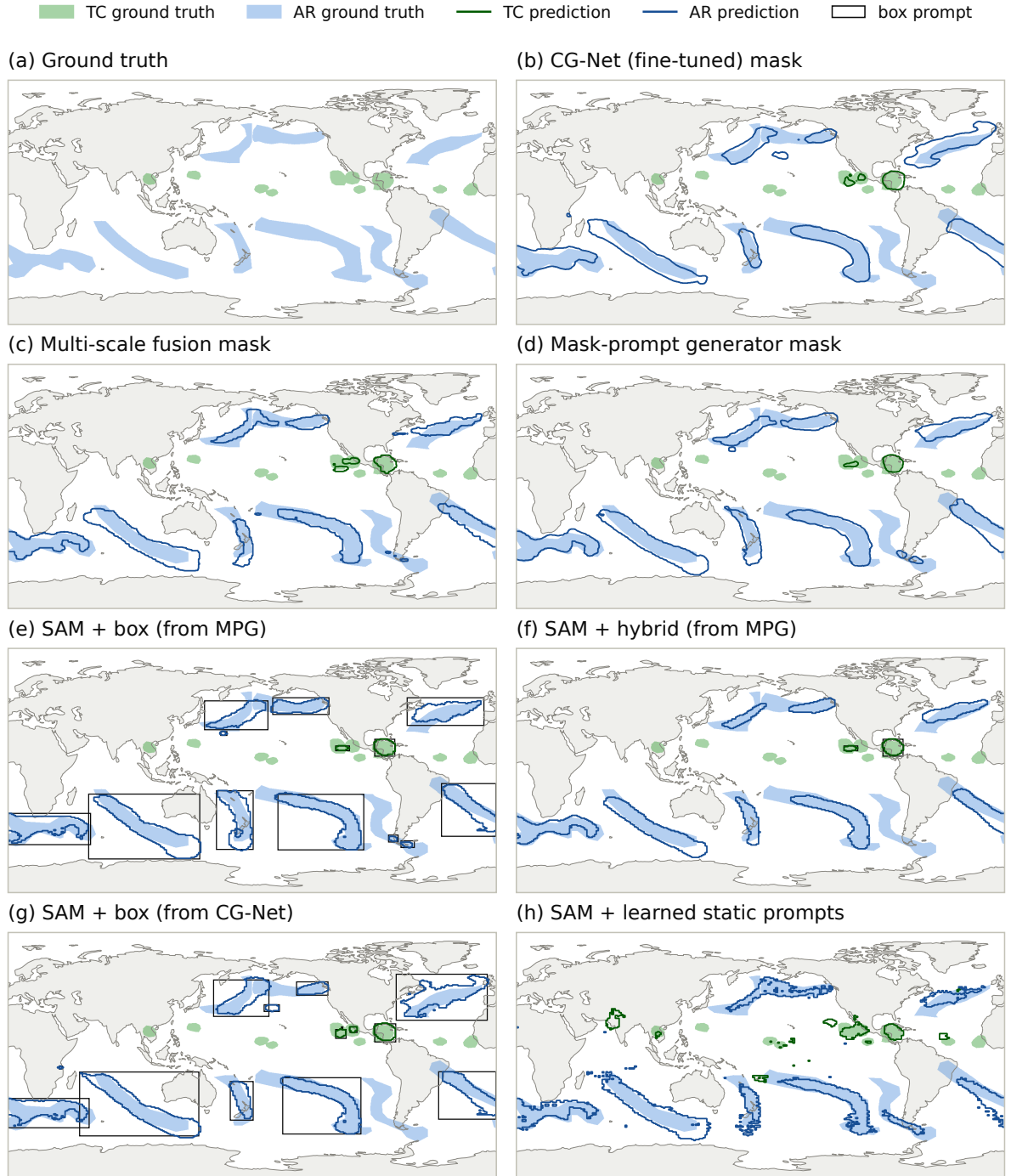


Figure A.1: Test image 45 (layout as in Figure 4.14).

Prompt type	Pos.	Neg.	Enlarge	IoU AR	IoU TC	IoU AR (orig. script)	IoU TC (orig. script)
CG-Net alone (baseline)				0.3817	0.3494		
point+bbox	10	5	0.0	0.3602	0.3447	0.3560	0.3430
bbox	0	0	0.0	0.3750	0.3423	0.3758	0.3404
point+bbox	15	10	0.0	0.3524	0.3420	0.3490	0.3399
bbox	0	0	-0.1	0.3327	0.3322	0.3355	0.3310
bbox	0	0	0.1	0.3625	0.3122	0.3572	0.3109
point	10	10	0.0	0.3376	0.3054	0.3297	0.3060
point	1	3	0.0	0.2856	0.3016	0.2859	0.3001
point	5	5	0.0	0.3425	0.2996	0.3380	0.3047
point	5	10	0.0	0.3402	0.2980	0.3342	0.3065
point	1	1	0.0	0.3224	0.2970	0.3214	0.2954
point	2	2	0.0	0.3260	0.2910	0.3226	0.2854
point	1	2	0.0	0.3013	0.2887	0.3009	0.2840
bbox	0	0	0.2	0.3343	0.2661	0.3260	0.2645
bbox	0	0	-0.2	0.2211	0.2494	0.2271	0.2523
bbox	0	0	0.3	0.3026	0.2279	0.2918	0.2268
bbox	0	0	0.5	0.2437	0.1597	0.2297	0.1613
point+bbox+mask	20	10	0.0	0.2352	0.1296	0.2469	0.1292
point+bbox+mask	10	5	0.0	0.2316	0.1129	0.2349	0.1022
mask	0	0	0.0	0.3794	0.0000	0.3724	0.0000

Table A.7: SAM with CG-Net as prompter, re-run on the 61 test images (Infused Token MLP 1.0, trained without label smoothing). “orig. script”: image passed through `ClimateSAM.set_infer_img`, which skips the learned input adapter (`test_prompt_effect.py`); other columns: input adapter applied as in training. Ordered by TC IoU.

A.2 Issues Found in the Original Phase-2 Code

1. `test_prompt_effect.py` encodes images with `ClimateSAM.set_infer_img()`, which feeds the first three raw channels to the encoder and skips the learned input adapter (`encode_images()` applies it). Effect: -0.00 to -0.03 IoU.
2. The multi-scale generators predict three channels, of which only TC and AR are supervised (as independent sigmoids) by `compute_generator_loss`; the untrained background channel takes part in the final argmax.
3. `LogisticRegressionPrompter` classifies the output of the first ViT block instead of the last.
4. `StreamSegMetrics.update(gt, pred)` is called with swapped arguments in several scripts. IoU is symmetric and unaffected; mean and frequency-weighted accuracies are affected.
5. All original Phase-2 training scripts select the best epoch on the test set.
6. The official ClimateNet CG-Net checkpoint predicts background everywhere in evaluation mode, because its stored BatchNorm running statistics do not match its weights; it works with batch statistics.
7. The fine-tuned CG-Net was trained on all 398 training images, so its outputs on training images are optimistic.